Who Qualifies?

Firm Responses and Selection into Preferential Tax Regimes *

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September 2026

Abstract

Preferential tax regimes can affect both the behavior of participating firms and which firms select into them when eligibility is endogenous. We study Brazil's 2018 Factor R reform, which introduced a large tax cut for eligible firms within the country's simplified turnover-tax regime. Eligibility required payroll to reach 28% of revenue, favoring labor-intensive firms while creating incentives to meet the threshold. We identify three margins of response. First, inframarginal firms receiving the tax cut raise revenues by 13%, through higher economic activity, without increasing employment. Second, initially ineligible firms adjust neither employment nor reported revenues, qualifying instead by shifting dividends to owner wages. Third, new registrations rise by 48%, attracting firms able to shift dividends to owner wages rather than those with greater employee labor intensity. Almost half of marginal entrants would otherwise have chosen an alternative tax regime; the rest are consistent with new business registrations. Preferential regimes can thus increase economic activity and registration without inducing the behavior targeted by their eligibility criteria. Endogenous eligibility shapes both firm behavior and selection, as firms adjust along the least costly margins. These results highlight a fundamental challenge in using endogenous firm characteristics to target tax incentives.

JEL Codes: D3, H2, H3, H5

*Corresponding author: Javier Feinmann (jfeinmann@berkeley.edu). We thank Gabriel Zucman, Damian Vergara, Jarkko Harju, Juan Carlos Suarez Serrato, Frederico Finan, Annette Alstadsæter, Theo Ribas Palomo, Davi Bhering, Juliana Londoño Velez, Lucie Gadenne, Camille Landais, Ségal Le Guern Herry, RFB Team and numerous seminar participants for insightful comments. We thank the Stone Center for financial support. All data work for this project involving confidential taxpayer information was done at the *Receita Federal do Brasil* (RFB) facilities, by RFB tax auditors. All results have been reviewed to ensure that no confidential information is disclosed. The views expressed in the paper do not necessarily represent those of the *Receita Federal do Brasil*, the World Bank and its affiliated organizations, the Executive Directors of the World Bank, or the governments they represent.

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1 Introduction

Small and medium-sized firms (SMEs) are central to developing economies, accounting for a large share of aggregate employment. Governments therefore implement numerous policies to encourage SME registration, growth, job creation, and other objectives. These policies often offer preferential treatment—such as tax cuts, subsidies, or simplified procedures—conditional on eligibility criteria that depend on firm behavior. Evaluating whether these policies achieve their intended goals is challenging as firms differ both in their responses to these incentives and in their ability to qualify for preferential treatment.

First, inframarginal firms that already meet the eligibility criteria automatically benefit from the new preferential treatment. Second, existing firms that do not qualify may find it profitable to adjust their behavior to become eligible. Their least costly way to qualify depends on their production technology, enforcement rules, and opportunities for avoidance or evasion. Third, the policy may encourage new firms to register to access preferential treatment, changing both the number and composition of registered firms.

These margins are difficult to distinguish empirically. Changes in eligibility and the effects of preferential treatment typically occur together, while the data needed to separate real activity from reporting responses and identify entrants' alternative choices are rarely available. This paper combines new Brazilian tax microdata with the 'Factor R' reform to disentangle these margins of firm response. In January 2018, the Brazilian government reformed *Simples Nacional*, the simplified tax regime at the core of the country's SME policy. The affected activities span a broad range of service-sector firms, from incorporated high-income self-employed workers to establishments with several employees. For firms in eligible economic activities, the reform made the turnover tax rate depend on the *payroll-to-revenue* ratio (Factor R). Payroll includes both employee compensation and owner wages. Firms with a Factor R of at least 28% faced a tax rate 9 percentage points lower than those below the threshold, creating a large tax notch. The reform thus made *Simples Nacional* more attractive to two types of firms: those whose existing labor intensity already met the threshold (*natural eligibility*), and those that could adjust their Factor R to qualify at a relatively low cost (*costly eligibility*).

The Factor R reform allows us to distinguish three margins of response. Among incumbent firms, we separately estimate the effects of the tax cut on firms initially above the threshold (High Factor R) and the adjustments made by firms initially below it (Low Factor R) to become eligible. For the latter, we distinguish changes in employment and reported revenues from shifts between dividends and owner wages. Finally, variation in eligibility across economic activities allows us to study entry into *Simples Nacional*, the types of firms that enter, and their counterfactual choices absent the reform. To identify

the different margins of response, we use *difference-in-differences* analyses comparing firms in eligible activities with firms in non-eligible activities that did not receive the tax cut.

Our first set of results concerns incumbent firms. We begin with High-Factor R firms, which automatically receive the turnover tax cut. Their reported revenues increase by 13% two years after the reform. This increase could reflect either reduced under-reporting or greater economic activity. To distinguish these channels, we combine electronic invoices for input purchases, credit card transactions, and bank account flows to construct a proxy for *true* revenues. We find that proxied economic activity also increases. However, employment remains unchanged, while variable capital increases significantly, suggesting that turnover taxation affects input choices unevenly.

We then turn to Low-Factor R firms, which must adjust their payroll-to-revenue ratio to qualify for the tax cut. These firms respond almost entirely by shifting dividends to owner wages, consistent with choosing the least costly adjustment margin. Owner wages rise gradually, reaching a 25% increase two years after the reform, while neither reported revenues nor employee payroll changes significantly. These adjustments contribute to a sharp bunching at the 28% threshold. However, this bunching may also reflect changes in the composition of entering firms, motivating our analysis of the extensive margin.

Thus, our second set of results concerns entry and selection into *Simples Nacional*. We follow cohorts of firms at the industry-code level rather than individual firms and find that the reform increased the number of new firms entering the regime in eligible sectors by 48%. To characterize the firms induced to enter, we decompose the ITT estimates into effects on inframarginal entrants (always-takers) and differences in the composition of marginal entrants (compliers). This allows us to assess whether selection reflects the labor intensity targeted by the policy or the ability to qualify through owner-income shifting. Marginal entrants have substantially higher owner wages relative to revenues, while their employee payroll-to-revenue ratio is essentially unchanged. Therefore, the reform disproportionately attracts firms using owner compensation to reach the 28% threshold, consistent with selection on opportunities for income shifting.

We then investigate the counterfactual choices of firms induced to enter *Simples Nacional*. We find no evidence of industry or labor contract relabeling.[1] Yet, entry into *Lucro Presumido*, the main alternative tax regime falls substantially in eligible industries. This decline accounts for 43% of the additional entry into *Simples Nacional*, with a similar pattern observed in firm stocks. The remaining share is consistent with new firm registration, although we cannot distinguish formalization of existing firms from business creation.

Finally, to compare the relative magnitudes of the different behavioral responses, we

[1] Labor contract relabeling, known in Brazil as *pejotização*, involves replacing an employment contract with a business-to-business contract to reduce social security contributions and personal income taxes.

translate them into a common monetary metric: their impact on government revenues. The extensive margin accounts for a substantial part of the policy's fiscal impact: revenue gains from additional registration offset about half of the mechanical revenue loss from the tax cut. Yet the contribution of new entrants depends critically on their counterfactual choices. Firms that would otherwise have entered the alternative tax regime generate revenue losses, substantially reducing the net fiscal gain from increased entry. Among incumbent firms, the margin of adjustment is equally consequential: revenue losses from income shifting are more than twice as large as revenue gains from increased reported revenues among High-Factor R firms.

Our findings highlight a central challenge in designing preferential tax regimes with behavior-based eligibility: firms adjust along the least costly margin, which may differ from the one policymakers intend to encourage. This shapes both incumbent responses and selection, attracting firms that qualify through income shifting rather than greater labor intensity. Evaluating these policies therefore requires examining how firms qualify, which firms are induced to enter, and their counterfactual choices absent the policy. Although our application concerns simplified tax regimes, widespread in developing countries, the mechanism extends to other policies with behavior-based eligibility, including R&D incentives, employment subsidies, and investment incentives.

Contributions to the literature

Our paper relates first to a broad literature in developing countries on firms' tax incentives, where preferential treatment is conditioned on criteria that firms can endogenously adjust. This literature documents a wide range of behavioral responses. Chen et al., 2021 shows that firms in China partly increase R&D investment, and partly relabel other investments to obtain a tax cut that depends on the ratio of R&D to revenues. Papers studying size-based policies show responses on revenues, costs, and even firm splitting to preserve preferential treatment (Onji, 2009; Best et al., 2015; Bachas and Soto, 2021; Lobel, Scot, and Zúniga, 2024; Nimier-David, Sraer, and Thesmar, 2026). We contribute by separately identifying how firms respond to the preferential tax regime itself, how existing firms adjust to meet the criteria when several margins are available, and how the extensive margin responses shape selection into the regime. In addition, we document that owners of SME firms are among the richest individuals in the country, highlighting who benefits from such policies (Agostini et al., 2018).

The extensive margin result connects our paper to a large development literature studying policies intended to bring small firms into the formal economy. Early evidence showed that reducing one-time registration costs, providing information, or simplifying registration often produces only modest increases in formalization, suggesting that for many

informal firms the private benefits of formality are insufficient to compensate for its recurring costs (De Mel, McKenzie, and Woodruff, 2013; Bruhn and McKenzie, 2014; De Andrade, Bruhn, and McKenzie, 2016; Ulyssea, 2020). Other interventions that increase the benefits of registration or substantially reduce ongoing tax and compliance costs can generate larger effects (Benhassine et al., 2018; Farias and Hsu Rocha, 2026). Importantly, however, additional registration does not mechanically imply firm growth, higher productivity, or greater tax revenue. We contribute to this literature by moving beyond the number of firms induced to register. We characterize the margin in which new firms induced by the policy select into the regime, and we study their counterfactual choices. This informs us about the underlying distribution of marginal entrants.

A growing strand of the public finance literature highlights that preferential regimes for SMEs have the potential to simultaneously increase formalization and reduce tax compliance, especially in settings with weak monitoring and enforcement (Bruhn and Loeprick, 2016; Wei and Wen, 2024; Hoy et al., 2024; Sharma et al., 2025; Gravoueille and Stantcheva, 2025). Our evidence strongly supports this theoretical work. The additional firm and revenue registration may come together with regime switching and tax avoidance behavior.

Our findings on owners increasing their own wages to receive the preferential treatment closely relate to the income-shifting literature. When labor and capital income face different tax treatments, owner-managers can respond to taxes by changing their payment scheme (Gordon and Slemrod, 1998). This margin has been particularly well documented in Nordic dual-income-tax systems, exploiting variation in dividend tax rates (Fjaerli, Lund, et al., 1997; Pirttilä and Selin, 2011; Le Maire and Schjerning, 2013; Harju and Matikka, 2014; Alstadsæter and Jacob, 2016; Tazhitdinova, 2020). Rather than the wedge between dividend and wage taxes, our paper exploits the notch on the payroll-to-revenue ratio, finding large income-shifting responses. We show that income shifting matters not only for existing firms, but it may be the margin in which potential entrants may select into the regime.

2 The Factor R Policy: setting and theory

Many policies offer firms preferential treatment conditional on satisfying an eligibility criterion. This is especially common in developing countries, where governments seek to promote the registration, growth, and employment of SMEs (e.g., tax benefits based on reported income, subsidies for employment creation, specific investments or activities, simplified regimes based on firm size, among others). Critically, when the criterion depends on endogenous firm behavior, the policy success is not straightforward. Firms that already satisfy the criterion receive the treatment without adjusting eligibility; firms that

do not may alter whichever component of the criterion is cheapest; and the policy may attract firms precisely because they face low costs of qualifying. Brazil's Factor R reform provides a setting in which these three forces can be separately identified.

2.1 Context

The simplified tax regime of *Simples Nacional*

We focus on the simplified tax regime of *Simples Nacional*, which constitutes the cornerstone of SME policy. It was established by Federal Law in 2006, and it covers a heterogeneous range of firms, from incorporated high-income self-employed to medium-sized labor-intensive firms. Table A.1 shows that it accounts for 80% of all firms and 31% of employment in Brazil. Firms in *Simples Nacional* are taxed on reported revenues, with their tax schedule depending on the economic activity, as they are presumed to have different profit margins (see Table A.2). SMEs often comprise a wide range of firms, especially in the service sector, the main focus of this paper. Table A.3 shows that about half of firms in the service sector can be categorized as self-employed who incorporate in a business.[2] The other half hire employees. Note that despite large differences in the tax schedules, the share of income after labor costs (proxy for profit margin) is very similar across service activities.

We also characterize the owners of firms in *Simples Nacional* as it determines the redistributive implications of policies addressing SMEs. Following the methodology in Palomo et al., 2025, we show that they rank between the 95th and 99.5th percentiles of the income distribution in the country.[3] This makes firms' behavioral responses central to the rationale for SME policies: since their owners are concentrated near the top of the income distribution, preferential treatment has little redistributive rationale on its own, placing greater weight on whether it induces the real responses these policies are intended to generate. We refer the reader to Appendix B for further details on the institutional context.

The Factor R Policy

On 1 January 2018, the Brazilian government introduced a reform that offers preferential tax treatment based on firms' payroll-to-revenue ratio (referred to as Factor R). Firms with a Factor R equal to or greater than 28% would pay revenue taxes under the Anexo III schedule (see Table A.2), while firms below the 28% threshold would pay at the Anexo

[2]The reader should think about them as high-income self-employed, as low-income self-employed are in the *Microempreendedor Individual* (MEI) regimes (revenue threshold of R$ 81,000). This is also a simplified tax regime with a flat payment, and it is not legally allowed to have more than one employee. It was designed to increase registration of informal employees (Farias and Hsu Rocha, 2026).

[3]We account for personal income as well as business income. We allocate the latter through a 10-layer shareholder tree.

V tax rate. Empirically, this translates into a 9 pp drop in the average revenue tax rate, representing a large tax notch in the system.[4]

Within *Simples Nacional*, only firms in specific service-related activities were eligible for the Factor R rule, providing a control group for the empirical analysis. However, within the eligible economic activities, the policy applied to both existing firms and new entrants, with the intended goal of encouraging the registration of labor-intensive firms, increasing employment at existing firms, and correcting the unequal tax burden that revenue taxation imposes on firms with different profit margins. Finally, the payroll considered in Factor R included not only employee payments but also owners' wages. Even if the overall impact on the government budget is negative (see Figure A.1), this may foster contributions to social security, insuring firm owners.[5] As a result, firms had several margins to qualify for the lower tax cut: reported revenues, employment, and shifting dividends to owners' wages.

2.2 Conceptual Framework for the Factor R Policy

Most governments design policies for SMEs with preferential treatment conditional on satisfying an eligibility criterion that depends on endogenous firm behavior. Such policies affect firms differently. Those who already meet the eligibility criteria receive preferential treatment right away. Existing firms that do not satisfy the criteria may find the preferential treatment profitable enough to adapt their behavior to reach the criteria. Finally, the preferential treatment may also induce new entrants as the regime becomes more attractive than the outside option. We provide a framework to conceptualize these margins through the Factor R case.

The Owner's Problem

Given the Factor R policy design, we can simplify the owner's problem as choosing an allocation $x_i = (L_i, e_i, \widetilde{R}_i, w_i^o)$, where L_i is hired labor, e_i is any other input[6], $\widetilde{R}_i$ is reported

[4] A graded version of this rule existed in the pre-reform Anexo V, covering a negligible number of firms (see Table J.12 in Online Appendix J).

[5] In Brazil, when the owner pays himself in wages (*pró-labore*), she must pay two wage taxes. The first is the contribution to the social security system (INSS) of about 11%. This payment is linked to social security benefits, such as retirement pension and life insurance for the family. The relative value of benefits over contributions explains why a significant share of firm owners pay themselves wages, even in the pre-policy period, when the dividend tax is zero. Figure A.1 shows that the tax notch is still substantial even after accounting for contributions. Second, owner wages are taxable income (different from dividends), and subject to the individual income tax (*Imposto de Renda sobre Pessoa Física*). However, the exemption threshold for the individual income tax is very large. Therefore, the level of owner wages that must be reported in order to offset the gains from the revenue tax cut is not binding for 95% of firms in Simples Nacional.

[6] For simplification, we treat e_i as a composite input, which can represent the owner's effort, capital, rent, etc.

revenues, and w_i^o represents owner income reported as wages.[7] Let the production technology be general and firm-specific, generating true revenues of the form $R_i = A_i F_i(L_i, e_i)$. A_i is a standard productivity shifter governing the firm's scale. Note that F_i is indexed by i, as firms may differ in several technological aspects, a common feature among SMEs. Firms with the same level of revenues can have substantially different labor shares, capital shares, profit margins, etc.

Let $\mathcal{T}_H(\widetilde{R})$ and $\mathcal{T}_L(\widetilde{R})$ denote total revenue-tax liabilities under the high and low tax schedule, respectively. Conditional on schedule $j \in \{H, L\}$, the owner's payoff is

$$\Pi_j(x_i) = R_i - \mathcal{T}_j(\widetilde{R}_i) - wL_i - v(e_i) - c(R_i - \widetilde{R}_i) - \Gamma(w_i^o) \tag{1}$$

The function $v(\cdot)$ captures the cost of composite input e_i.[8] The function $c(\cdot)$ is the cost of underreporting revenue, and $\Gamma(\cdot)$ is the cost of reporting owner income as wages rather than dividends. The latter includes payroll contributions and personal income taxes, as well as tax-planning costs and the risk of penalties. Then, let $r_i \equiv \frac{wL_i + w_i^o}{\widetilde{R}_i}$; the reform introduced the following change in the tax schedule:

$$\text{Pre-reform:} \quad \mathcal{T}_j(\widetilde{R}_i) = \mathcal{T}_H(\widetilde{R}_i)$$

$$\text{Post-reform:} \quad \mathcal{T}_j(\widetilde{R}_i) = \begin{cases} \mathcal{T}_H(\widetilde{R}_i) & \text{if } r_i < T, \quad T = 0.28, \\ \mathcal{T}_L(\widetilde{R}_i) & \text{if } r_i \geq T, \quad T = 0.28. \end{cases}$$

Thus, the reform does not simply lower the revenue tax. It makes access to the lower schedule conditional on an endogenous statistic that firms can alter through hired labor, owner compensation, reported revenue, or changes in production.

The Value of the Reform

Since preferential treatment depends on firms' endogenous behavior, the owner's maximum payoff can be used to derive the reform's value. Let's define the owner's maximum payoff under the unconstrained high tax regime, unconstrained low tax regime, and the constrained low tax regime:

[7] We assume firms take wages as given, which is likely to be true for SMEs. In any case, in the empirical analysis we will look at employees' payroll, covering any possible wage adjustment.

[8] For example, when e_i represents owner effort, this includes the opportunity or disutility cost of the owner's time.

$$\Pi_H \equiv \max_{x_i} u_H(e_i, L_i, \widetilde{R}_i, w_i^o)$$

$$\Pi_L \equiv \max_{x_i} u_L(e_i, L_i, \widetilde{R}_i, w_i^o)$$

$$\Pi_T \equiv \max_{x_i : r(x_i) \geq T} u_L(e_i, L_i, \widetilde{R}_i, w_i^o)$$

Thus, we can decompose the value of the reform into two components. First, the unconstrained value of a lower tax schedule:

$$B \equiv \Pi_L - \Pi_H, \quad B > 0 \ \ \forall i \ \text{ with } \ R_i > 0 \tag{2}$$

Second, the cost of reaching the eligibility rule, comparing the allocation the firm would ideally choose under the favorable tax schedule with the best allocation available under that same schedule once the Factor R restriction is imposed:

$$C \equiv \Pi_L - \Pi_T \geq 0. \tag{3}$$

Importantly, C is not the cost of any particular behavioral response: the firm can re-optimize jointly over hired labor, the composite input e_i, owner compensation, and revenue reporting.

As a result, the value of the Factor R reform for a given firm is determined by:

$$\begin{aligned} G &\equiv \max\{\Pi_H, \Pi_T\} - \Pi_H \\ &= [B - C]_+ . \end{aligned} \tag{4}$$

Equation (4) is the central object of the framework. The reform increases the value of *Simples Nacional* for a firm when the benefit of receiving the favorable tax schedule exceeds the minimum loss required to satisfy the eligibility rule.

Natural vs. Costly Eligibility

Equation (4) shows that two groups of firms benefit from the policy. First, firms whose low-tax unconstrained allocation already meets the eligibility criteria. We call these firms *naturally eligible*. This is the case for sufficiently labor-intensive firms with high payroll-to-revenue ratios. Interestingly, naturally eligible firms are useful for isolating firms' behavioral responses to a revenue tax cut.

Second, firms whose low-tax unconstrained allocation does not meet the criteria, but who can meet it at a low cost ($C < B$). We call these firms *costly eligible*. As these firms optimally choose the cheapest adjustment margin, their responses depend on their production

technology, enforcement rules, avoidance and evasion opportunities, etc. Costly eligible firms under the Factor R reform must adjust either employees' payroll, owner wages, or reported revenues to meet the 28% threshold, satisfying:

$$w\Delta L_i + \Delta w_i^o - T\Delta \widetilde{R}_i = (T - r_i^0)\widetilde{R}_i^0 > 0. \tag{5}$$

Finally, Proposition 1 in Appendix C derives the conditions that characterize which firms are more likely to be naturally and costly eligible. This is helpful for understanding who the compliers are and determining who benefits from the preferential treatment.

Selection into the Regime

The value of the reform (G) does not affect only existing firms in the eligible economic activities covered by the policy. It also affects selection into the regime, based on the same principles of natural and costly eligibility. Labor-intensive firms may now find it beneficial to opt for *Simples Nacional*. In addition, firms that can meet the 28% threshold cheaply enough may also find registration in the regime attractive.

We consider potential firms choosing among *Simples Nacional* (S), *Lucro Presumido* (P), and an outside option (0). *Lucro Presumido* is the closest alternative tax regime available for these economic activities, in which firms are taxed on presumed profits (another way of implementing a revenue tax). The key difference is that it is not a simplified tax regime, meaning that firms must file state and other taxes separately, increasing the compliance costs. We consider an outside option that entails not registering the firm's economic activity. This may include inactivity as well as informality, as we cannot empirically distinguish between them.[9] Critically, the reform changed the value of opting in for *Simples Nacional* relative to its alternatives. Let K_S denote the fixed cost of operating under Simples Nacional, including registration and compliance costs. Pre-reform, the value of choosing S for a given firm was $V_S^0 = \Pi_H - K_S$, while after the reform:

$$\begin{aligned} V_S^1 &= \max\{\Pi_H, \Pi_T\} - K_S \\ &= V_S^0 + G. \end{aligned} \tag{6}$$

Equation (6) shows that Factor R raises the value of selecting into S by exactly the same object G that governs the incentive to obtain the favorable schedule for firms already operating in the regime. The implication is that Factor R makes S more valuable for both

[9] Empirically, we will consider several other counterfactuals such as relabeling the economic activity for firms within *Simples Nacional*, and switching from employment to B2B contracts. As we find that none of them is relevant, except for *Lucro Presumido*, we stick to the simplest framework.

firms that naturally satisfy the eligibility requirement and firms that can meet the 28% threshold cheaply enough. Whether this increase in value actually induces a firm to select into S depends on its alternative regime choice.

An additional feature of the selection margin is what the most likely counterfactual is for different firms. On the one hand, inducing firm registration ($O - S$ boundary) is a very common policy goal, and there is a large literature in development economics studying how to foster registration. On the other hand, inducing firms to switch tax regimes just for a tax advantage ($S - P$ boundary) is likely to be an unintended policy goal. Proposition 3 in Appendix C derives the conditions that characterize which firms are more likely to come from each possible counterfactual.

3 Data and Empirical Analysis

Having established the potential behavioral responses induced by the Factor R policy at the intensive and extensive margin, we proceed to explain how we estimate them empirically.

3.1 Data

This paper uses detailed Brazilian tax data that were previously unavailable for research purposes. This is possible thanks to a collaboration between researchers at the International Tax Observatory (ITO) and *Receita Federal de Brasil* (RFB). Given the stringent confidentiality policies, all data processing has been conducted on RFB's premises by a tax auditor. We work with the following databases:

PGDAS:
It is the monthly tax declaration and payment platform for firms in the simplified tax regime of *Simples Nacional*. Each record is tied to a firm identifier (CNPJ) and a reference month. Firms self-report their Anexo, monthly revenues, and 12-month accumulated revenues. If firms are subject to the Factor R rule, then they self-report the monthly total payroll and 12-month accumulated payroll, used to compute their Factor R coefficient.

GFIP:
It is an electronic reporting system in which employers are required to report labor and tax obligations. Each record is linked to an employer identifier (CNPJ), a reference month, and a "payment reason" (*rubrica*). This data is a perfect complement to PGDAS for two reasons. First, it contains payroll information for all firms in all periods, regardless of whether they

are subject to the Factor R rule. This allows us to measure the Factor R coefficient for non-eligible firms, as well as pre-policy values for eligible firms. Second, because it records the payment reason, we can directly distinguish between employees' payroll and owners' wages, which is crucial for measuring alternative behavioral responses. Importantly, the PGDAS manual explicitly instructs the use of GFIP's payroll for reporting payroll in PGDAS when eligible for the Factor R. Consequently, we construct the Factor R value for every firm in every period using GFIP information. For firms that report both, Figure A.2 shows that they closely track each other.

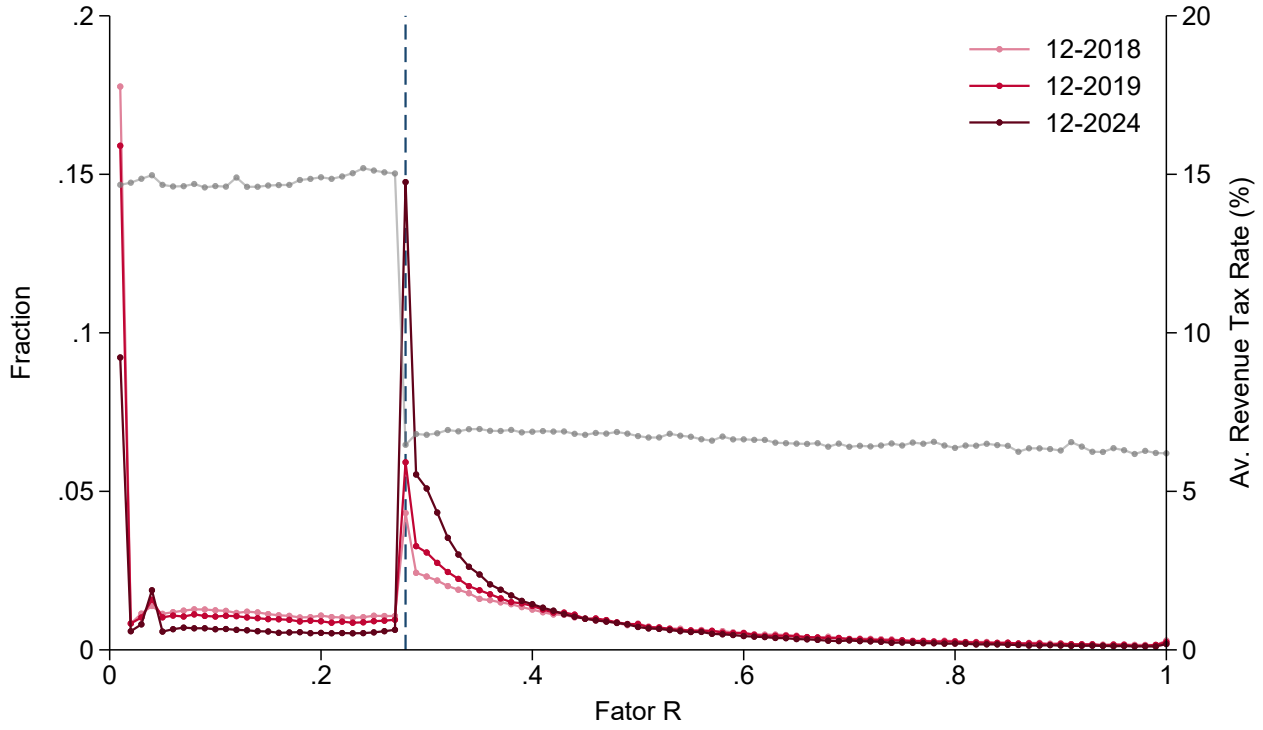
Other Databases:
We use many other data sources for specific tasks. We use ECF, the tax form for firms in Lucro Presumido, to measure whether firms switch tax regimes. We also leverage information on electronic invoices, employment records, credit card consumption, and bank account flows to proxy for real economic activity when interpreting the effects on firms' reported revenues. We combine these variables using a stochastic production frontier model developed by the tax authority (RFB, 2023), aiming to infer the firm's *true* revenues (see Online Appendix I for details). Finally, we also use **DIRPF**, the individual income tax form, which allows us to characterize firms' owners and look at their individual tax obligations.

3.2 Empirical Analysis

As explained earlier, the policy states that eligible firms with a *payroll-to-revenue* ratio (Factor R) of 28% or higher face a large drop in their revenue tax rate, averaging 9 pp, introducing a massive tax notch in the system. Consistently, Figure 1 documents a large bunching response with about 15% of all firms subject to Factor R reporting a payroll-to-revenue ratio exactly at 28%.[10] Moreover, there is an interesting dynamics in the bunching pattern that persistently grows year after year, which could be caused by either incumbent firms learning over time or by new entrants.

[10]While tax notch theory predicts this behavior (Kleven, 2016), most empirical analyses fail to find such a clean pattern and attribute the difference to adjustment frictions (Bachas and Soto, 2021; Lobel, Scot, and Zúniga, 2024).

Figure 1: Factor R distribution over time



Notes: This figure shows both the probability density function of the Factor R variable (left axis) and the discontinuity in revenue tax rates (right axis). Red lines show the fraction of all firms subject to the Factor R rule at each possible Factor R value. Each red line represents a different time period. The gray line plots the mean of the average revenue tax rate paid by firms in each Factor R bin. The blue dashed line indicates the 28% threshold. Bins are of .01 size.

Importantly, standard bunching techniques are insufficient to learn about the different margins of firm's responses for several reasons. First, the bunching may reflect how firms adjust the Factor R coefficient, but it is silent about *how* they do it (through which component). Moreover, identifying the marginal bunching requires the incentives to be uniform, which is unlikely to hold in this setting,[11] as for a given payroll-to-revenue ratio, firms differ in incentives due to their scale (see Appendix C).

Therefore, we will exploit both the industry-eligibility criteria and the timing of the policy to conduct several *difference-in-differences* (DiD) analysis. The core of the empirical analysis is Equation 7, where i indexes the unit of analysis and t the time period relative to the reform.

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}[\text{Treated}_i = 1]\mathbf{1}[t = k] + u_{it}, \tag{7}$$

where α is a unit fixed effect, and λ a period fixed effect. However, we separate our analysis into the intensive margin (incumbent firms in *Simples Nacional*) and the extensive margin. This allows us to separately identify all the behavioral responses from Section 2.2.

[11] Pollinger, 2025 makes this point explicit, including the case of measuring extensive margin responses.

For incumbent firms, the Factor R policy generated two clean quasi-experiments. Firms that had a pre-policy Factor R above 28% (High FR) were automatically eligible for the tax cut, allowing us to identify how firms respond to the preferential tax treatment. Firms that had a pre-policy Factor R below 28% (Low FR) had to increase their payroll-to-revenue ratio to obtain the preferential treatment by increasing employment, reducing reported revenues, or shifting dividends to owner wages. Therefore, the incumbent firms analysis is split by Low and High FR firms.

One way to motivate this sample splitting is to connect it with the conceptual framework of natural and costly eligibility. Proposition 2 in Appendix C derives such connection. We use, as a first approximation, that behavioral responses of High FR firms are mostly due to naturally eligible firms, while behavioral responses of Low FR firms are driven by costly eligible firms. We validate this interpretation in Appendix D, where we estimate the effect on the entire Factor R distribution for incumbent firms.

For the extensive margin, the unit of analysis is different. If firms select into *Simples Nacional* due to the policy, looking at new entrants is central object. Unlike incumbents, new entrants cannot be observed every period. Therefore, we conduct the analysis at the industry code, comparing cohorts of new entrants in those economic activities that become eligible with those that did not.[12][13] Then, we can estimate the effect on the stock and new firms. Moreover, we recover an important parameter of interest: how new entrants induced by the policy differ from new firms that would have entered anyway. This allows us to characterize the underlying motive for entering and connect it back to natural vs. costly eligibility. Finally, we also estimate firms' counterfactual.

4 Intensive Margin

This section focuses on how incumbent firms (those in *Simples Nacional*) respond to the Factor R policy and what we can learn from such behavior. Our sample comprises firms with positive revenues in all quarters between 2016 and 2019, covering two years before and after the reform. We define our treatment group as firms belonging to Anexo VI prior to the reform, and our control group as those belonging to Anexo IV prior to the reform. The choice of Anexo IV as the control group is intentional. Table A.2 shows that firms in Anexo IV are, within the service sector, the ones facing the lowest tax rate and, therefore, with no incentives to relabel their economic activity to take advantage of the Factor R

[12] More precisely, we use the 7-digit CNAE codes, and we map them with the economic activities covered by Anexo III, IV, V, and VI pre-policy.

[13] For robustness, we also perform a data-driven approach to define different groups of exposed and non-exposed industries. We select industry codes that have a large share of all firms in Anexo VI for the treated group, and the analogous ones for the control group. Results are virtually identical.

policy.[14] Table A.4 presents the different activities for treatment and control. Both treated and control correspond to firms in the service sector. The activities in the treated group were all automatically subject to the Factor R rule, while those in the control group were not. Table 1 illustrates the baseline differences between the treatment and control group.

Table 1: Descriptive Statistics: Firms in 2017-Q4

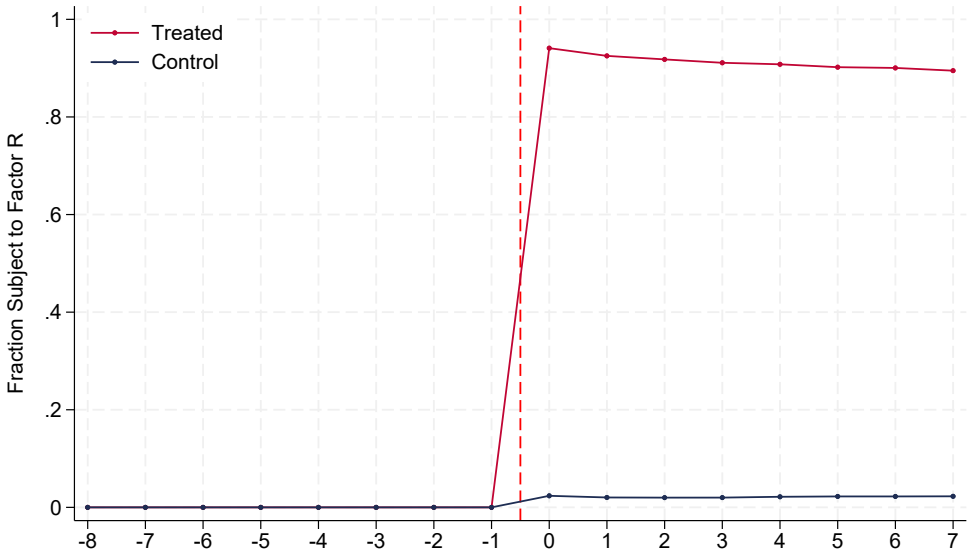
	Accumulated 12-month				Current (Quarter)			
	Low FR		High FR		Low FR		High FR	
	Treated	Control	Treated	Control	Treated	Control	Treated	Control
Revenues	308,565	543,037	220,681	513,755	28,403	51,261	20,575	47,432
Owner Wages	17,526	17,673	33,878	29,774	1,558	1,558	2,857	2,475
Employees' Payroll	23,927	38,690	70,277	220,735	2,564	4,145	7,226	22,383
$\mathbf{1}$[Revenues = 0]	0	0	0	0	0	0	0	0
$\mathbf{1}$[Owner Wages = 0]	0.278	0.347	0.092	0.128	0.298	0.371	0.105	0.150
$\mathbf{1}$[Employees' Payroll = 0]	0.581	0.476	0.269	0.100	0.603	0.512	0.289	0.122
$\mathbf{1}$[Total Payroll = 0]	0.182	0.221	0.000	0.000	0.198	0.244	0.003	0.004
N Firms	19,863	29,594	16,320	17,692	19,863	29,594	16,320	17,692

Notes: This table reports descriptive statistics for firms observed in 2017-Q4. Low FR and High FR refer to firms classified according to their pre-reform Factor R. Treated firms are those affected by the reform, while control firms belong to the comparison group. Monetary values are in Brazilian reais.

Figure 2 shows how the treatment and control groups are differently exposed to the policy. This confirms that almost all firms in the treated group became subject to the Factor R rule, while only a negligible fraction of firms in the control group did, probably due to natural change in economic activity.

[14] In any case, we find no evidence of industry relabeling and robustness exercises show that changing the control group generates no qualitative difference in the results (see Figure A.6

Figure 2: Fraction subject to Factor R by pre-policy Anexo



Notes: This Figure shows the fraction of firms subject to the Factor R policy separately by treatment status. The red line shows the group of firms that operated in Anexo VI pre-reform, while the blue line shows those that operated under Anexo IV pre-reform. The x-axis is the quarter relative to the reform, where $t = 0$ represents the first quarter of 2018.

A central feature is that the Factor R policy generated two clean quasi-experiments. Firms initially above the 28% threshold became automatically eligible for a lower revenue tax rate, whereas firms initially below the threshold had to change their payroll-to-revenue ratio to obtain the tax reduction. Therefore, we split the analysis based on pre-policy Factor R in order to learn not only how firms adjust their behavior to obtain preferential treatment, but also how firms directly respond to it.

Results: Behavioral Responses by pre-reform Factor R group

We begin by showing the effect on the average revenue tax rate. Figure 3 clearly shows how the policy affected incentives differently depending on the location in the Factor R distribution pre-policy. Those who were above 28% (High FR) face an immediate drop in the revenue tax rate of about 9pp. Those who were below 28% (Low FR) do not face a large and sudden drop, but rather an initially smaller drop which grows over time. This is due to the fact that Low FR firms do not become automatically eligible, but more and more firms enjoy the tax cut as they cross the threshold.

Revenues

We study how the policy affected reported revenues. Figure 3b shows null effects for Low FR firms, indicating that alternative margins are cheaper to increase the payroll-to-

revenue. However, High-FR firms see a significant increase in reported revenue of about 10% eight quarters after the reform. We refer the reader to Appendix E, where we translate the reduced-form estimate to a revenue elasticity to the net-of-tax rate.[15]

The increase in reported revenues for High-FR firms is consistent with re-optimization. However, two alternative stories are consistent with responding to a revenue tax cut. A firm can either increase true economic activity and, therefore, revenues, or they can reduce revenue under-reporting. Below, we explore in more detail what drives the increase in reported revenues.

Employees' Payroll

We now turn to the effect on employees' payroll. As Equation 5 shows, this is an alternative for Low FR firms to reach the 28% threshold. Any adjustment in the number of employees or their wages will be reflected here. Similarly, employers could decide to register informal employees or reduce payments under the table within the firm (Feinmann, Hsu Rocha, and Lauletta, 2022). We may also expect an increase in employees' payroll for High FR firms if the increase in reported revenues is due to real activity, and the non-deductibility of labor costs was restraining labor demand considerably.

Figure 3c shows basically no effects on employees' payroll for either group. Neither Low FR nor High FR firms experienced increases statistically different from zero. For the first group, this reveals that adjusting employees' payroll is not the cheapest option to take advantage of the tax rate reduction, undermining the policy goal. The result for the High FR group is also surprising. Despite the fact that revenue taxation distorts input demand and that we find a 10% increase in reported revenues, employees' payroll does not respond to the policy. This reveals that firms are more reactive in other margins.

Owner Wages

The other important behavioral response is how firm owners adjust their payments between dividends and wages when it can affect their overall tax rate. Given that Low FR firms neither reduced reported revenues nor increased employees' payroll, Equation 5 implies that owners must pay themselves in wages to increase the payroll-to-revenue ratio and obtain the tax cut. For firms above 28% pre-policy, the responses depend on whether they belong to the constrained or unconstrained group, and whether the former group finds owner wages the cheapest way to relax the constraint.

Figure 3d shows large and growing increases in owner wages for Low FR firms, reaching 25% two years after the reform. Interestingly, High FR firms also exhibit a significant

[15]Such elasticity assumes no constrained firms because we cannot measure the right change in the revenue tax rate for them, as we cannot isolate the shadow price of the constraint. We still consider it helpful to provide an order-of-magnitude estimate for such elasticity.

increase in owner wages. As mentioned earlier, this may be explained by a considerable mass of constrained firms above the threshold. Figure A.5 in Appendix A.1 shows evidence that firms right above the threshold explain most of the owner wages increase, while firms far from the threshold explain most of the revenue increase.[16]

The comparison between the behavioral responses in owner wages and employees' payroll reveals how cheap it is for firms in simplified tax regimes to exploit loopholes. This is an important result for thinking about the proper design of such regimes, as exemptions, differential schemes, and discretionary rules may create loopholes that undermine policy goals.

Figure 3: Reduced Form Estimates for Incumbent Firms



(a) Average revenue tax rate

(b) Annual Revenues

(c) Employees' Payroll

(d) Owner Wages

Notes: This figure shows the reduced form coefficients ($\beta_k s$) from Equation 7 where the outcomes are the average tax rate (Panel a), the 12-month accumulated revenues (Panel b), the 12-month accumulated employees' payroll (Panel c), and the 12-month accumulated owner wages (Panel d). The outcome variable is in levels in all cases. Accumulated 12-month outcomes are measured up to 4 periods pre-reform, as it is consistent with incumbents' definition (two years before and after the reform). For the average revenue tax rate, we run an OLS specification, while for Factor R components we use Poisson regressions as the distribution contains many zeros. The blue line limits the sample to firms with a Factor R below 28% pre-policy. The red line limits the sample to firms with a Factor R equal to or greater than 28% pre-policy. Vertical bars show 95% confidence intervals.

[16] An alternative explanation (not modeled in our framework) could be that owners use some fixed rules for self-payments (e.g. 20% wages 80% dividends). If so, the increase in owner wages for High FR firms may be mechanically related to the increase in revenues. Consistently, Figure A.4d shows that the ratio of owner wages to revenues remains totally flat for High FR firms.

Reporting vs. Real Responses for High FR Firms

One striking result is the significant revenue response together with the null employment response for High-FR firms. This finding raises a natural question: what explains the increase in firms' revenues? To answer this question, we rely on a *production frontier analysis*. This exercise uses several variables related to firms' real economic activity to predict *true* revenues (see Section 3.1 for details). We further decompose the effect on alternative inputs.

Table 2 shows that firms' revenues estimated through predictors of true economic activity also increase, and almost in the same magnitude as reported revenues. Consistently, we observe that variable capital adjusts upward strongly after the policy. It is worth noting that the pre-period coefficient on true revenues is negative and significant, indicating a small violation of the parallel trend assumption. While the post coefficients are almost identical to those for reported revenues, we are cautious in our interpretation and do not claim that the entire effect on revenues is real, as part of it may be explained by pretrends. However, part of it seems to be real, and the types of inputs adjusting are consistent with that.

Table 2: Reported vs. True Responses

	(1) Reported Revenues	(2) True Revenues	(3) Fixed Capital	(4) Variable Capital	(5) Current Purchases
Pre-period					
2016	-0.008	-0.033**	0.002	0.021	-0.038
	(0.023)	(0.016)	(0.010)	(0.024)	(0.025)
Post-period					
2018	0.045***	0.046***	0.009	0.053***	0.019
	(0.012)	(0.013)	(0.009)	(0.015)	(0.019)
2019	0.083***	0.082***	0.023*	0.065***	0.024
	(0.020)	(0.018)	(0.013)	(0.021)	(0.024)
FE by Firm	✓	✓	✓	✓	✓
FE by Year	✓	✓	✓	✓	✓
Observations	135,864	128,228	128,228	128,228	128,228

Notes: This table presents the results of estimating Equation 7 on a series of outcomes. Column (1) shows the effect on the annual reported revenues (last quarter). Column (2) uses the predicted revenues based on a series of variables related to real economic activity (see Section 3.1 for details). Columns (3), (4), and (5) show the estimates for fixed and variable capital, and current purchases, respectively. All specifications are estimated using PPML and include firm and year fixed effects. The difference in observations is explained by 5.6% of firms for which we were not able to reconstruct their input variables. Standard errors are clustered at the pre-reform industry level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5 Extensive Margin

This section focuses on the extensive margin responses to the policy. In particular, we are interested in understanding whether the policy induced more entry, why they entered, and what their counterfactual would have been in the absence of the policy. As explained in Section 3.2, the unit of analysis is different. We estimate Equation 7 at the industry level, following cohorts of firms over time for exposed and non-exposed industries.

Before proceeding with the results, it is worth addressing a methodological challenge. If firms select into *Simples Nacional* due to the policy, looking at cohorts of new entrants is a central object. One of the parameters of interest is how new firms induced by the policy differ from new firms that would have entered regardless of the Factor R reform. This informs us about the underlying motives to enter the regime. A standard composition analysis (estimating Equation 7 for endogenous outcomes such as Factor R, employment, etc.) confounds our parameter of interest with the treatment effect on the firms that would enter even in the absence of the policy. Therefore, we next explain how to address such a challenge, and we then present the results.

Marginal Entrants vs. Inframarginal Behavioral Responses

As we are interested in understanding the selection of new firms induced by the policy, it is important to interpret β_k precisely. After the reform, there are two groups of new firms: the ones that entered because of the policy (*marginal entrants* or *compliers*) and those that would have entered anyway (*inframarginal entrants* or *always-takers*). While the latter group is inframarginal to the entry decision, they may change their behavior under the Factor R rule. This means that β_k captures a combination of the selection of new firms and the behavioral responses of always-treated firms. Formally:

$$\beta^{DiD} = \underbrace{\Big(\mathbb{E}[Y(1) \mid AT] - E[Y(0) \mid AT]\Big)}_{\text{(I)}} + \underbrace{\pi_C\Big(\mathbb{E}[Y(1) \mid C] - \mathbb{E}[Y(1) \mid AT]\Big)}_{\text{(II)}}. \tag{8}$$

Term (I) is the intensive-margin response of firms that would have entered regardless. Term (II) is the contribution of the composition effect to the reduced-form effect. Dividing it by the complier share (π_C) delivers our main object of interest: the selection gap. It reveals how those new entrants induced by the policy differ from those firm that would have entered anyway. Their differences in behavior and characteristics shed light on the entry motives. See Online Appendix H for details on the derivation of Equation 8.

Estimating π_C is straightforward, as it comes from using the log of total new firms as the outcome variable. However, recovering the selection gap requires knowledge of the

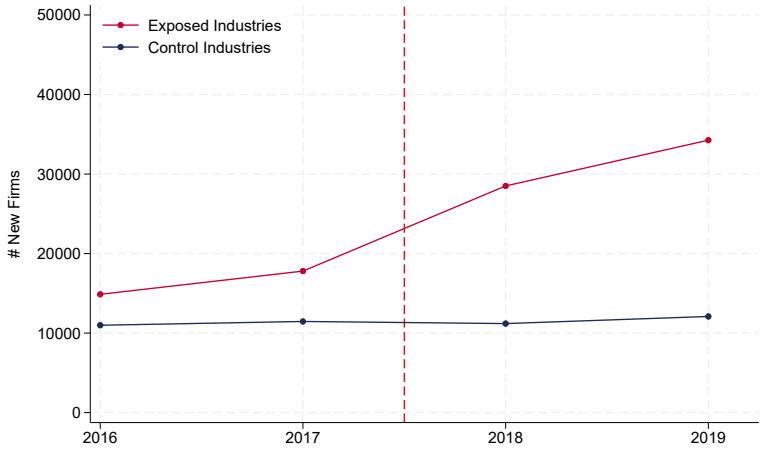
change in behavioral responses among inframarginal entrants. Note that this is a setting with large changes in incentives, so we expect firms to respond accordingly. Also note that Term (I) cannot be estimated, as always-takers are not directly observed in the data. Fortunately, in Section 4 we estimated the behavioral responses of incumbent firms, which can be used to approximate how inframarginal entrants respond to the policy. Concretely, we assume that, absent the reform, the Low- versus High-FR composition of inframarginal entrants would match that observed among pre-reform entrants, and that conditional on this status their behavioral response equals the corresponding incumbent response.[17] For outcomes that are predetermined with respect to the policy, we instead set $\Delta_{AT} = 0$.

Results: Entry, motives and counterfactual

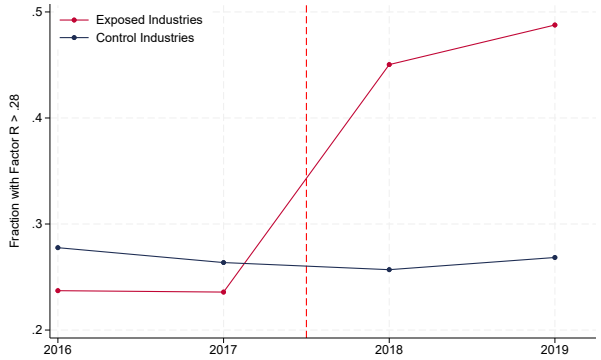
We begin by providing visual evidence of the change in the number of new firms and their composition for exposed and non-exposed industries. We then estimate Equation 7 at the industry level. Figure 4 confirms that the effects are driven by changes in the treated rather than in the control group.

[17] This is not a theoretical assumption, but rather an empirical assumption. We do not observe how inframarginal new firms would have behaved in the absence of the policy, but we rely on our incumbents' responses as a proxy.

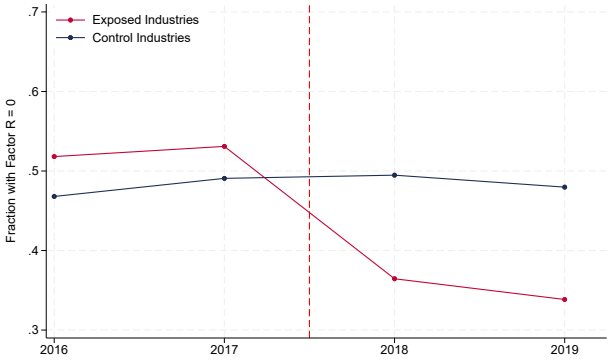
Figure 4: Raw trends for new firms (exposed vs. non-exposed industries)



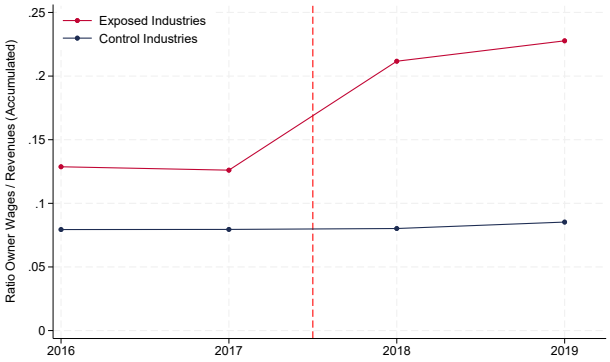
(a) Number of New Firms



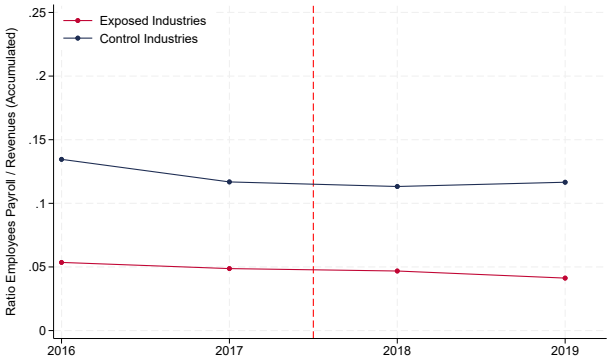
(b) Firms with FR ≥ 28%



(c) Firms with FR < 5%



(d) Avg. Ratio Owner Wages / Revenues



(e) Avg. Ratio Employees' Payroll / Revenues

Notes: This Figure shows the raw trends comparing treated vs control industries. The unit of observation is the industry-year, constructed from information on new firms within each industry-year. Panel (a) plots the evolution of the total number of new firms. The remaining plots show the evolution of the new firms' composition. Panel (b) plots the share of new firms that entered with a Factor R above 28%. Panel (c) plots the share of new firms that entered with a Factor R below 5%. Panel (d) and (e) decompose the Factor R by each component of the numerator: owner wages and employees' payroll, respectively. The vertical dashed bar indicates the introduction of the reform. Revenues, owner wages, and employees' payroll were previously expressed in real terms. Averages across industries are weighted by each industry's relative size.

Table 3 presents the *difference-in-differences* coefficients. Columns (1) and (2) show large effects in the number of new firms in exposed industries post-reform, both in levels and

in logs. This recovers the number of new firms due to the policy, from which we derive the share of always-takers and compliers (π_{AT} and π_C, respectively)

Columns (3), (4) and (5) estimate the β_k^{DiD}s for Factor R-related outcomes to show that the cohorts of new firms in the exposed industries changed relative to the non-exposed industries in a way that is fully consistent with the policy. New firms in treated industries become 20-24pp more likely to have a Factor R above 28%, which basically doubles the baseline likelihood. More than half locate right above the threshold and come from the lowest part of the Factor R distribution. The latter result helps rationalize Figure 1, where a considerable share of bunchers seems to come from the missing mass at zero Factor R. While incumbent bunchers were more likely to come from right below the threshold, the new entry of firms explains the large missing mass at the lower part of the Factor R distribution.

New Entry and Decomposition

Table 3: Extensive-Margin Difference-in-Differences Estimates

	(1) # New Firms	(2) Log New Firms	(3) Pr(FR > .28)	(4) Pr(.28 < FR < .4)	(5) Pr(FR < .05)	(6) Owner Wages/Revenues	(7) Employees Payroll/Revenues
Pre-period							
2016	-25.244**	-0.112	-0.010	-0.008*	0.008	0.002	-0.011**
	(10.820)	(0.092)	(0.011)	(0.005)	(0.014)	(0.005)	(0.005)
Post-period							
2018	129.444***	0.379***	0.209***	0.107***	-0.161***	0.081***	0.000
	(31.568)	(0.078)	(0.014)	(0.009)	(0.014)	(0.006)	(0.005)
2019	180.055***	0.452***	0.240***	0.131***	-0.175***	0.093***	-0.006
	(46.114)	(0.069)	(0.018)	(0.014)	(0.014)	(0.007)	(0.004)
Mean Dep. Var. Treated	206.95	4.15	0.24	0.06	0.53	0.13	0.05
Mean Dep. Var. Control	208.47	4.05	0.26	0.07	0.49	0.08	0.11
FE by Year	✓	✓	✓	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓	✓	✓	✓
Reweighted	X	X	✓	✓	✓	✓	✓
Observations	564	472	466	466	466	466	466

Notes: This table reports event-study difference-in-differences estimates of the Factor R reform on new entry. Each column corresponds to a separate regression. Coefficients are interactions of year indicators with treatment exposure, defined at the industry level, with 2017 as the omitted reference year. All specifications absorb year and industry (CNAE) fixed effects. Standard errors clustered at the industry level in parentheses. *Mean Dep. Var.* rows report the 2017 baseline mean of the dependent variable for treated and control groups. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Columns (6) and (7) estimate the β_k^{DiD}s for the different components of Factor R. This is the first step in identifying what motivates the new entry of firms. We find that new firms almost doubled the share of owner wages over revenues, but they do not change employees' payroll over revenues at all. We now turn to tease apart how much of this effect is explained by infra-marginal firms changing their behavior due to the policy, versus

the composition of the new firms induced by the policy, following Equation 8.

As explained above, we recover term (I) using the behavioral responses of incumbent firms estimated in Section 4, weighting the estimates by the pre-policy share of new entrants located below and above the 28% threshold. Table 4 then presents the decomposition of β^{DiD}, together with the resulting selection gap and the comparison between marginal and no-reform entrants across several outcomes.

Table 4: Behavioral and Compositional Decomposition

	Pr(FR > 0.28)		Owner W/R		Employee P/R	
	2018	2019	2018	2019	2018	2019
Total effect, β^{DiD}	0.209	0.240	0.081	0.093	0.000	-0.006
Behavioral component, (I)	0.066	0.109	0.019	0.027	-0.001	-0.001
Composition component, (II)	0.143	0.131	0.061	0.066	0.002	-0.005
Composition share	69%	55%	76%	71%	–	–
$E[Y(1) \mid C] - E[Y(1) \mid AT]$	0.456	0.402	0.196	0.204	0.005	-0.014
$E[Y(1) \mid C] - E[Y(0) \mid AT]$	0.522	0.511	0.215	0.231	0.004	-0.015

Notes: AT denotes firms that would have entered *Simples Nacional* without the reform; C denotes induced entrants. Component (I) imputes the response of AT using the incumbent estimates, and component (II) is the remaining composition effect. The share of induced entrants, π_C, is the detrended increase in entry divided by observed entry in treated industries. The selection gap compares marginal and inframarginal entrants under the same post-reform environment (fifth row). The marginal vs. no-reform gap additionally incorporates the behavioral response of inframarginal entrants and compares marginal entrants with the entrants that would have been observed absent the reform (sixth row). Composition outcomes are weighted by CNAE size in 2017

Consistent with the policy incentives, we see that, even when only about a quarter of pre-policy entrants had a Factor R above 28%, the selection gap is very large. Compliers are 45pp more likely to enter with a Factor R above 28% than inframarginal entrants, even after accounting for their behavioral responses. Moreover, marginal entrants are no different from inframarginals in terms of labor intensity. Rather, marginal firms have a 20 pp larger owner wages-to-revenue ratio than inframarginal firms.

What motivates marginal entry? Our conceptual framework identifies two mechanisms through which Factor R can raise the value of Simples. Firms may benefit because their unconstrained choices naturally generate a sufficiently high Factor R, which is more likely for sufficiently labor-biased technologies, or because they can achieve eligibility at low cost. Empirically, we find no evidence that marginal entrants are selected on hired-

labor intensity. Instead, their distinguishing Factor R component is owner compensation, consistent with selection on the income-shifting margin.

Counterfactual

Understanding what marginal entrants would have done in the absence of the reform is central to evaluating its extensive-margin effects. Proposition 3 in Appendix C provides a productivity ordering of these counterfactuals: conditional on the non-scale technology parameter (F_i), firms induced from *Lucro Presumido* are higher-productivity than firms induced from the outside option. Empirically, we therefore examine whether increased entry into Simples is accompanied by displacement from alternative formal tax regimes or instead by an increase in firms coming from inactivity/informality.

Because the single pre-reform coefficient is small but not exactly zero, we net out a linear extrapolation of the 2016–2017 differential trend from the post-reform coefficients. All extensive-margin magnitudes reported in the text (the entry effect, the Lucro Presumido offset, the complier share πC, and the revenue terms in Section 6) use these trend-adjusted coefficients; the unadjusted coefficients are reported in Tables 3, 5, and A.5, and the adjustment matters little for the qualitative conclusions.

Table 5 shows the difference-in-differences result for exposed and non-exposed industries regarding new firms by tax regime. By 2019, the reform increased entry into Simples Nacional by approximately 48 percent relative to its estimated no-reform level. The associated decline in entry under Lucro Presumido offsets about 43 percent of that increase.

Table 5: Extensive-Margin: New Firms Entry

	(1) Subject to FR	(2) # Simples Nacional	(3) # Lucro Presumido	(4) # Total
Pre-period				
2016	0.001	-25.244**	-3.550	-28.794**
	(0.006)	(10.820)	(4.672)	(13.138)
Post-period				
2018	0.565***	129.444***	-40.079***	89.365***
	(0.022)	(31.568)	(11.910)	(23.927)
2019	0.543***	180.055***	-48.497***	131.558***
	(0.025)	(46.114)	(17.018)	(35.782)
Mean Dep. Var. Treated	0.009	206.953	149.814	356.767
Mean Dep. Var. Control	0.000	208.473	13.255	221.727
FE by Year	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓
Observations	472	564	564	564

Notes: This table shows the effects of the policy on new entrants. The sample comprises cohorts of new entrants for exposed and non-exposed industries. Column (1) shows the likelihood of being subject to the Factor R rule. Column (2) shows the effects on the number of new firms in *Simples Nacional*, while Column (3) uses the number of new firms in *Lucro Presumido*. Column (4) calculates the net effect on new firms. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

While the substitution of new entrants in one regime for another is prevalent and straightforward, it is not the only possible counterfactual. Incumbent firms in Lucro Presumido may switch to Simples Nacional (showing up as new entrants), firms close to graduating from Simples Nacional may prefer to not to do so, and existing firms in Simples Nacional may become less likely to close. Although we cannot tease apart these alternative channels, Table A.5 shows the changes in stock resulting from the policy, which summarizes all such channels. Accounting for the small differential trend in the stock of firms in Simples Nacional, Lucro Presumido accounts for half of the increase in the stock of firms in Simples Nacional.

Alternative counterfactual

The firm entry effects could be explained by other adjustment margins. One alternative is relabeling of industry code. It is possible that firms performing economic activities that became subject to the Factor R rule were registered under another economic activity to avoid the high tax schedule. If this is true, the increase in the number of new firms in the exposed industries should be offset by a reduction in the number of firms in other

industry codes, presumably those relatively similar. Figure A.6 shows that the policy did not reduce the number of new firms in any subgroup of economic activities related to Anexo III pre-policy. Similarly, Figure 4a shows a flat evolution in the control group, suggesting no offsetting from Anexo IV.

Another potential counterfactual is the so-called *pejotização*. This is a widespread practice in Brazil, where employees are hired through business-to-business contracts, rather than standard employment. This allows the employer to avoid payroll taxes, and the employee to face a lower tax burden. It requires the employee to open an incorporated firm to receive the payments. The Factor R reform made this practice more profitable, as employees could pay themselves 28% in wages and face a lower revenue tax rate. We test whether the policy induced more *pejotizacao* by looking at the likelihood that the owners of the new firms were previously working as employees. Columns (3), (4), and (5) in Table A.6 show that the cohorts of new firms did not increase the share of owners coming from employment contracts.

6 Comparing the Behavioral Responses

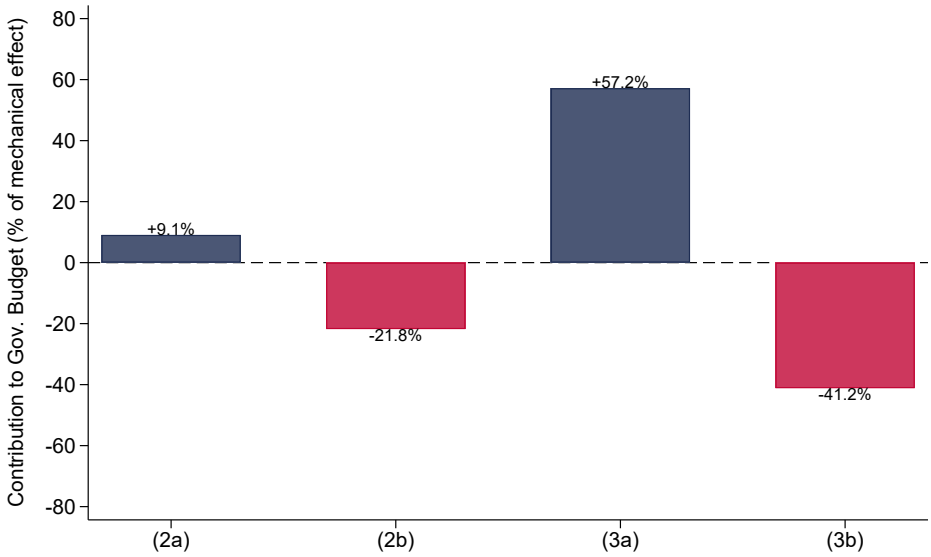
We have shown that incumbent firms facing a revenue tax cut increase reported revenues, and while employment does not move, part of the increase seems to be a real response. Then, we have also shown that incumbent firms find income shifting the cheapest way to achieve preferential tax treatment, compared to employment or revenue responses. On the extensive margin, we find an increase in the number of new firms registered, driven by the income-shifting vehicle, with part of this increase crowding out the alternative tax regime. This section aims to translate these different responses into a common monetary metric to compare their relative orders of magnitude. We calculate the contribution of each behavioral response to the government budget. We use the policy's mechanical effect as the benchmark.

$$\Delta\text{Gov. Budget} = \underbrace{-(\tau_{VI} - \tau_{III})\, R^{hi}}_{(1)} + \underbrace{\tau_{III}\, \hat{\rho}\, R^{hi}}_{(2a)} - \underbrace{(\tau_{V} - \tau^{P}_{III})\, N^{lo}\, \hat{m}}_{(2b)} + \underbrace{\tau^{P}_{III}\, \bar{R}_0\, \hat{N}_0}_{(3a)} - \underbrace{(\tau_{LP} - \tau^{P}_{III})\, \bar{R}_{LP}\, \hat{N}_{LP}}_{(3b)} \tag{9}$$

Equation (9) has five terms. Term (1) captures the *mechanical* effect of reducing the revenue tax rate to incumbents with pre-policy Factor R above 28%, which exists regardless of any behavioral response. Terms (2a) and (2b) account for the contribution of incumbents' behavioral responses to the government budget. Term (2a) enters positively because high-FR firms increase the reported revenues in the share ρ. Term (2b) enters negatively because a revenue-weighted share of firms crosses above the threshold through income shifting, resulting in a lower revenue tax rate. Finally, terms (3a) and (3b) measure the contribution

of the entry margin to the government budget. Note that (3a) enters positively, as a share of the new firms would not have registered in the absence of the policy. While we cannot differentiate whether it is new business activity or formalization, both interpretations have equal effects on the government's budget. Term (3b) enters negatively because the remaining share of firm entrants would have operated under the alternative *Lucro Presumido* tax regime in the absence of the policy, paying the corresponding taxes. Appendix F provides a detailed explanation of how each term is calculated. We take into account behavioral responses and compliers' characteristics at each margin, as highlighted in the conceptual framework.

Figure 5: Comparison of Behavioral Margins



Notes: This figure shows the relative contribution of each behavioral response in Equation 9 to the government's budget. The mechanical effect, totaling R$ 572 million, is used as the benchmark. (2a) accounts for the increase in reported revenues of High-FR incumbent firms. (2b) accounts for the reduction in tax payments by Low-FR incumbent firms engaged in income shifting. (3a) accounts for the additional firms registering their revenues, while (3b) accounts for the crowding out of *Lucro Presumido*. For the extensive-margin responses, we report the linear pretrend-adjusted estimates. The red and blue bars represent negative and positive effects on the government's budget, respectively. See Appendix F for details.

As expected, the policy's mechanical effect is large. Cutting revenue taxes by about 9pp for eligible firms with Factor R above 28% represents an annual loss of R$ 572 million for the government. Figure 5 shows that the behavioral responses are also sizable. From the incumbents' side, the income-shifting channel had a larger effect on the government's budget than the increase in reported revenues. The extensive margin plays an even more important role, and understanding firms' counterfactuals is critical. Not accounting for the fact that a large share of new entrants would have been in an alternative tax regime may lead us to overestimate the registration effects.

A general lesson is that behavioral responses are sizable for SMEs, but they may not all

align with policy goals or even with the direction of their impact on revenue. Evaluations that fail to consider the alternative ways that existing firms can adjust or the extensive margin responses would miss a very important part of the policy effect. Furthermore, looking at firms' entry without differentiating counterfactuals can also be misleading.

7 Conclusions

This paper studies how firms respond and select into preferential policies when benefits depend on characteristics that firms can themselves adjust. Brazil's Factor R reform provides unusual variation to study both margins. The reform sharply reduced revenue tax rates for firms with payroll above 28 percent of revenues, increasing the value of Simples Nacional both for firms that were naturally labor intensive and for firms that could reach the threshold by changing their behavior.

The evidence reveals a range of behavioral responses for firms differently affected. First, incumbent firms that were already above the threshold received the tax cut automatically. This led to substantial increases in reported revenues, at least partly associated with real activity. When turning into inputs, we find an increase in variable capital, but no effect on employment. Second, existing firms initially below the threshold have to increase their payroll-to-revenue ratio to receive the tax cut. They responded almost entirely by shifting owner income from dividends to wages, rather than by increasing employee payroll or reducing reported revenue. Among the alternatives, income shifting seems to be the least expensive adjustment margin. There is a wide range of tax avoidance or evasion strategies available for SMEs, which typically interact with policies' design (e.g. relabeling, under-reporting, informality, etc.), and our results confirm that SMEs can cheaply make use of them.

The extensive margin responses turned out to be very important as well. The reform increased entry into Simples Nacional by about 50 percent. The marginal entrants were no more intensive in hired labor. Instead, they were disproportionately firms that used owner compensation to satisfy Factor R. Therefore, the underlying distribution of potential entrants does not include relatively more labor-intensive firms but a substantial mass of firms willing to register under a lower tax burden. Moreover, around 43 percent of the additional flow into Simples Nacional displaced firms that would otherwise have entered another formal tax regime. Large take-up therefore need not indicate a correspondingly large effect on the economic activity a policy is designed to favor.

More broadly, our results highlight two forces that matter for the design of preferential policies. The real behavior policymakers seek to affect may itself be relatively inelastic, and alternative ways of qualifying for a benefit may be considerably cheaper. Heterogene-

ity in these adjustment costs then affects not only the behavior of existing beneficiaries but also the composition of those induced to participate. Evaluating such policies therefore requires looking beyond take-up and statutory eligibility measures to understand both how firms obtain preferential treatment and who responds to the policy.

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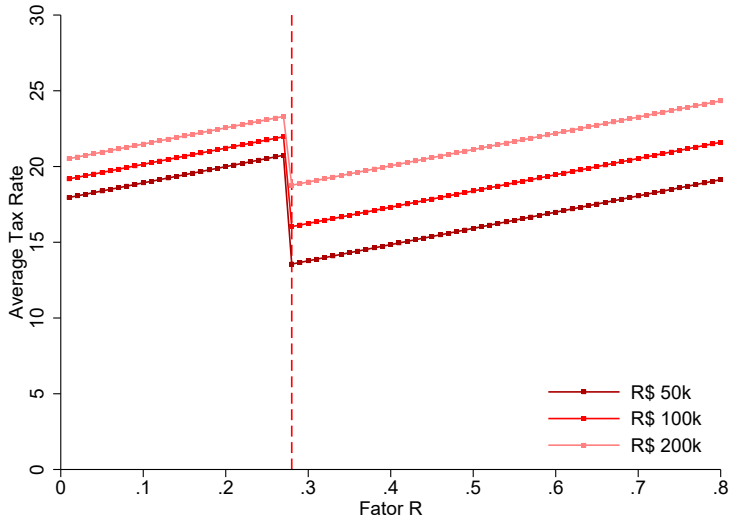
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A Additional Figures and Tables

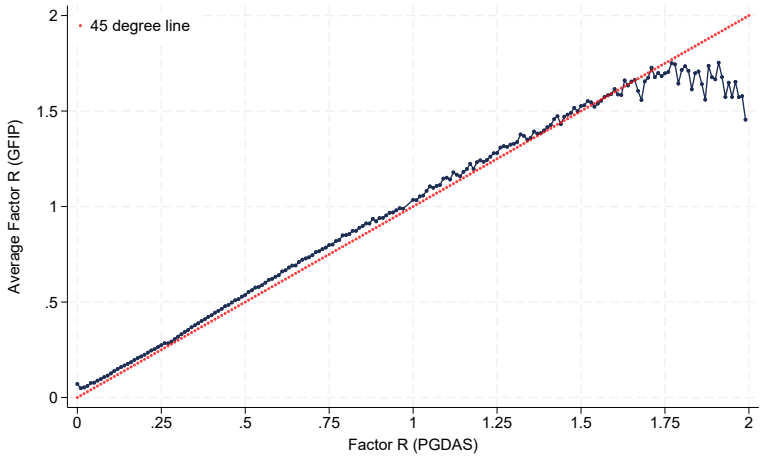
A.1 Figures

Figure A.1: Change in Effective Tax Rate including All Wage Taxes



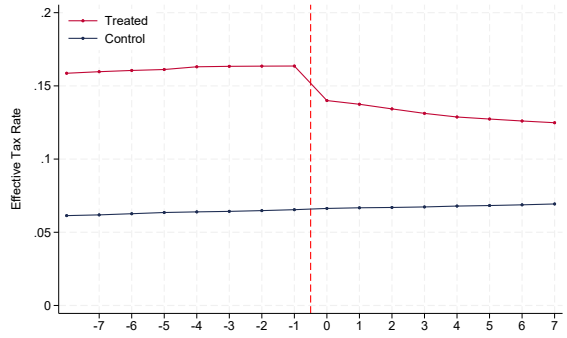
Notes: This Figure plots the average tax rate (average revenue tax + INSS) across the Factor R distribution, for different revenue levels.

Figure A.2: Factor R in PGDAS vs. GFIP

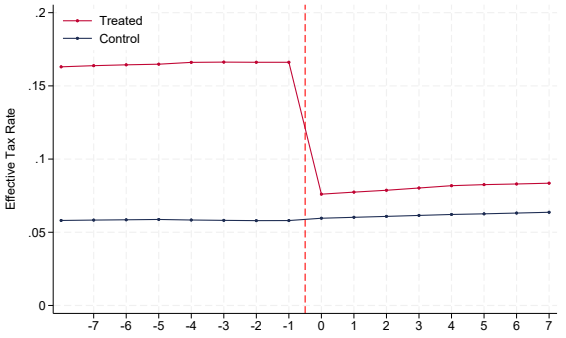


Notes: This figure compares the Factor R calculated from the tax forms (PGDAS) and from the electronic reporting system (GFIP). The sample comprises firms that are mandated to report both. Factor R bins in PGDAS are of size .01. The red lines shows the 45 degree line.

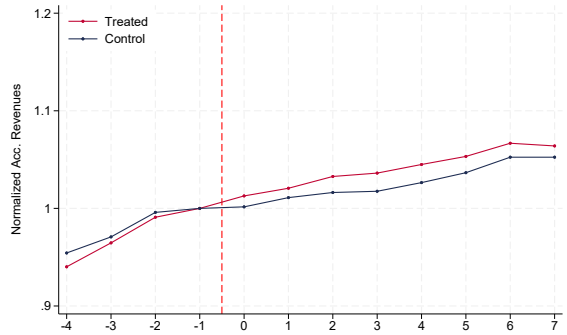
Figure A.3: Treatment vs. Control: Raw trends for incumbent firms



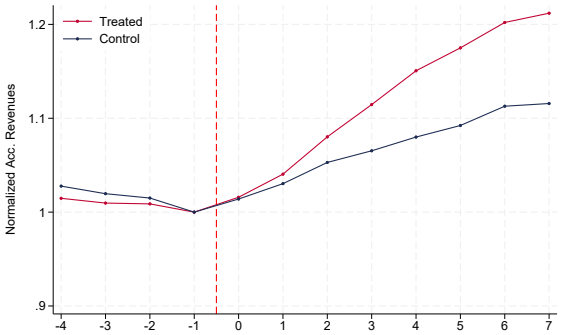
(a) Av. Revenue Tax Rate - Low FR



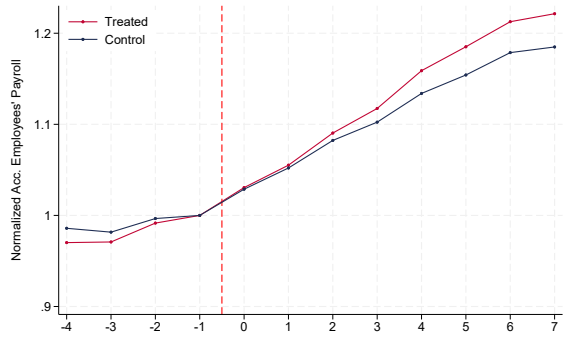
(b) Av. Revenue Tax Rate - High FR



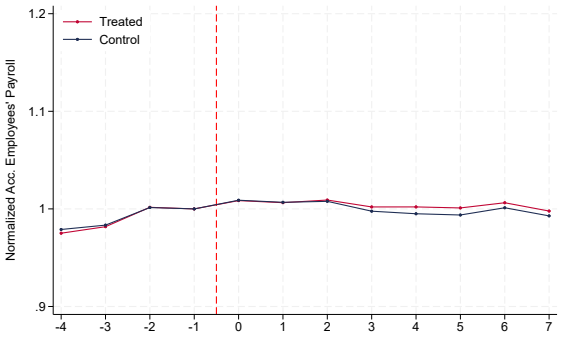
(c) Annual Revenues - Low FR



(d) Annual Revenues - High FR



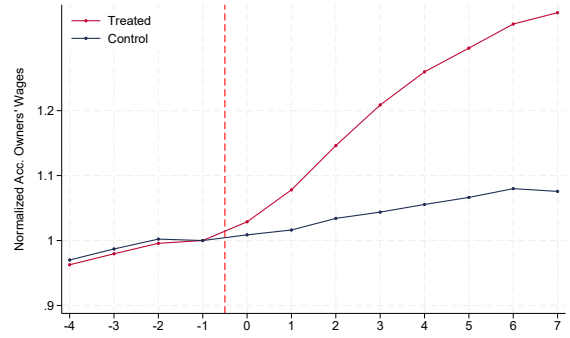
(e) Employees' Payroll - Low FR



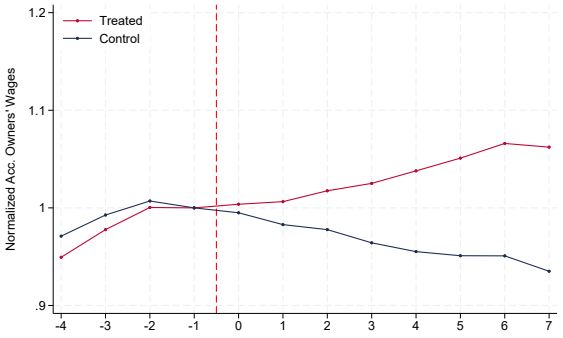
(f) Employees' Payroll - High FR

Notes: This Figure shows the raw trends comparing treated vs control incumbent firms. Panels on the left correspond to Low FR firms, while panels on the right correspond to High Factor R firms. Panel (a) and (b) plot the evolution of the average revenue tax rate. Panel (c) and (d) plot the evolution of the average revenues, normalized in the last quarter of 2017. Panel (e) and (f) plot the evolution of the average employees' payroll, normalized in the last quarter of 2017. The vertical dashed bar indicates the introduction of the reform. Revenues and employees' payroll were previously expressed in real terms.

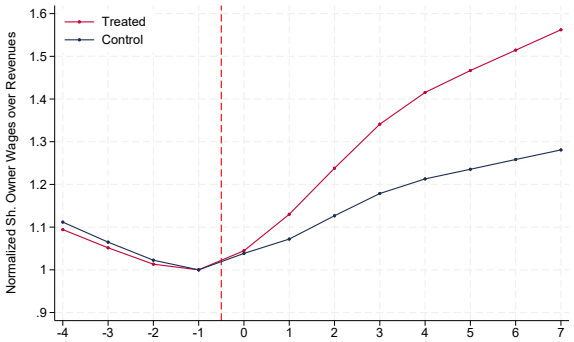
Figure A.4: Treatment vs. Control: Raw trends for incumbent firms (more outcomes)



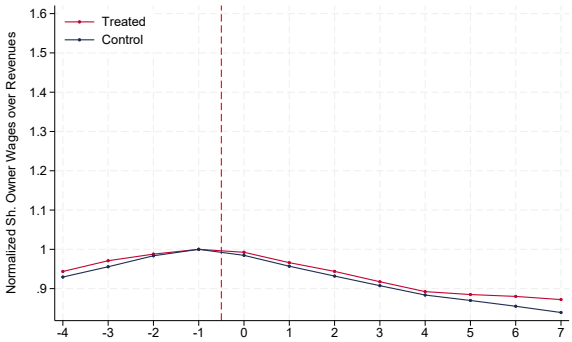
(a) Owner Wages - Low FR



(b) Owner Wages - High FR



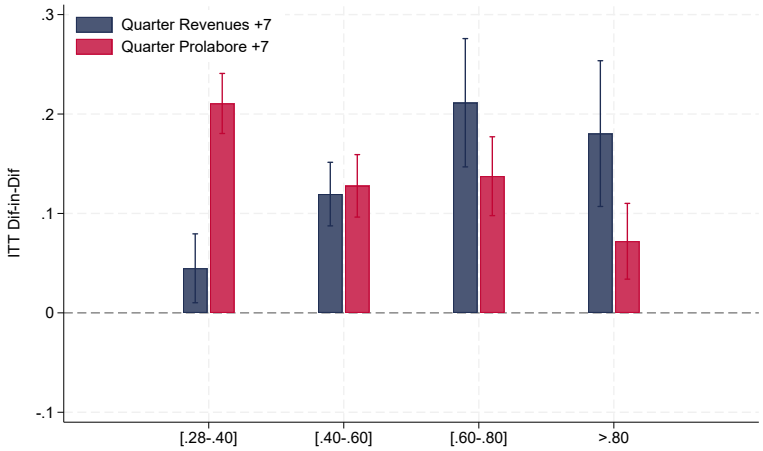
(c) Owner Wages / Revenues - Low FR



(d) Owner Wages / Revenues - High FR

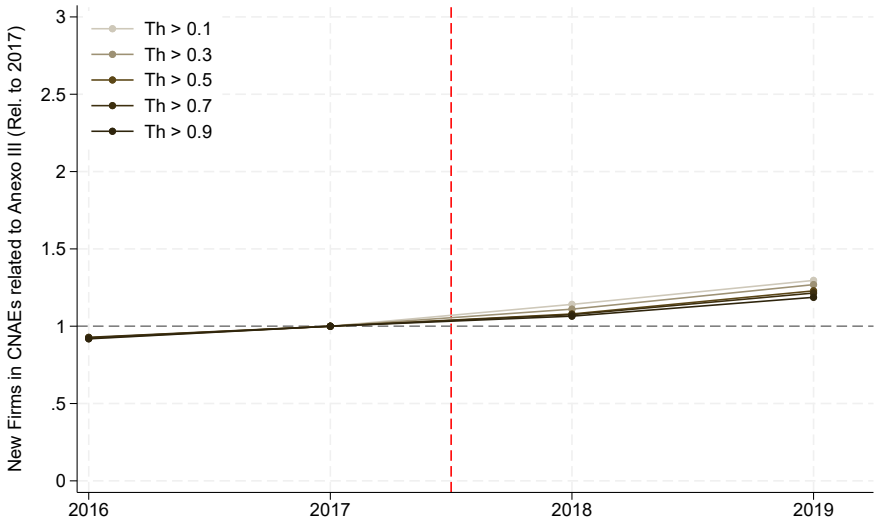
Notes: This Figure shows the raw trends comparing treated vs control incumbent firms. Panels on the left correspond to Low FR firms, while panels on the right correspond to High Factor R firms. Panel (a) and (b) plot the evolution of the average wages paid to the firm's owners, normalized in the last quarter of 2017. Panel (c) and (d) plot the evolution of the share of revenues used to pay owner wages, normalized in the last quarter of 2017. The vertical dashed bar indicates the introduction of the reform. Revenues and owner wages were previously expressed in real terms

Figure A.5: ITT for Incumbent Firms based on pre-policy Factor R



Notes: This Figure shows the β_7 from Equation 7 for different samples based on the pre-policy location in the Factor R distribution. Blue bars use deflated reported revenues as outcome, and red bars use owner wages. From left to right, the sample moves further away from the 28% threshold.

Figure A.6: New firms related to Anexo III



Notes: This Figure shows the evolution in the number of new firms for the economic activities related to Anexo III. Each line represents a different threshold to define the group of industry codes. For example, the lightest uses the industry codes for which at least 10% of the firms pre-policy were listed in Anexo III. The darkest only considers those industry codes for which at least 90% of the firms were listed in Anexo III. The red dashed line indicates the policy event.

A.2 Tables

Table A.1: Totals by Tax Regime (2022)

	Firms	Tax Payments (in millions R$)	Employees	Payroll (in millions R$)
Lucro Real	220,384	1,354,174	34,126,577	1,584,977
Lucro Presumido	1,073,033	201,994	7,745,465	290,290
Simples Nacional	5,560,139	150,618	19,000,890	660,637
MEI	9,429,910	253,050	5,185	2,483
Totals	16,283,466	1,711,972	61,134,749	2,538,389
Totals – MEI	6,853,556	1,706,787	60,872,933	2,535,905
Share Simples (excl. MEI)	81.13%	8.08%	31.21%	26.05%

Notes: This table reports general statistics regarding the Brazilian tax system. The rows refer to the alternative tax regimes, and the columns show their relative importance in terms of number of firms, tax payments, employment, and payroll.

Table A.2: Tax Schedule by Anexo (2018-present)

Annual Revenues (R$)	**Anexo I**	**Anexo II**	**Anexo III**	**Anexo IV**	**Anexo V**
≤ 180,000	4.00%	4.50%	6.00%	4.50%	15.50%
180,000 to 360,000	7.30%	7.80%	11.20%	9.00%	18.00%
360,000 to 720,000	9.50%	10.00%	13.50%	10.20%	19.50%
720,000 to 1,800,000	10.70%	11.20%	16.00%	14.00%	20.50%
1,800,000 to 3,600,000	14.30%	14.70%	21.00%	22.00%	23.00%
3,600,000 to 4,800,000	19.00%	30.00%	33.00%	33.00%	30.50%

Notes: This table reports the different tax schedules in *Simples Nacional*. Anexo I corresponds to retail, Anexo II to manufacturing, and Anexo III, IV, and V to different service activities.

Table A.3: Summary Statistics by Anexo (2019)

	Anexo 1	Anexo 2	Anexo 3	Anexo 4	Anexo 5
Av. Annual Revenues	447,681	680,658	340,617	423,092	237,294
Med. Annual Revenues	207,520	317,244	147,170	177,558	106,264
Share self-employed (%)	28.8	18.7	47.6	48.7	56.8
Share with Employees (%)	71.2	81.3	52.4	51.3	43.2
Av. Employees Payroll (incl. zeros)	62,245	165,501	74,451	79,824	44,107
Proxy Profit Margin (%)	86.1	75.7	78.1	81.1	81.4
Number of Active Firms	1,186,799	173,554	1,174,837	157,077	240,110

Notes: The table reports summary statistics separately by Simples Nacional Anexo. Annual revenues and employee payroll are expressed in nominal Brazilian reais. Self-employed firms are defined as firms reporting zero employee payroll. The proxy profit margin is computed as one minus the ratio of average employee payroll to average annual revenues

Table A.4: Broad economic activities by Anexo (Simples Nacional)

Control: Anexo IV	**Treated: Anexo VI**
Construction (incl. subcontracting), interior design and landscaping services	Health professionals (medical, dental, veterinary; psychology and related clinics)
Security, cleaning, and maintenance services	Regulated professional services (architecture/engineering; technical consulting and auditing)
Legal services (advocacy)	Other intellectual services (e.g. advertising/journalism; translation/interpreting; appraisals)

Notes: This table provides illustrative examples of activities classified under Anexo IV (control) and Anexo VI (treated) prior to the reform. Categories are based on LC 123/2006, art. 18, §5º-C and §5º-I, and are shown for exposition rather than as an exhaustive list.

Table A.5: Extensive-Margin: Stock of Firms

	(1) Subject to FR	(2) # Simples Nacional	(3) # Lucro Presumido	(4) # Total
Pre-period				
2016	-0.001	-141.914	14.671	-127.243
	(0.001)	(158.246)	(51.056)	(173.791)
Post-period				
2018	0.534***	466.549**	-207.034***	259.515
	(0.022)	(187.235)	(53.573)	(170.653)
2019	0.526***	839.358**	-303.281***	536.077
	(0.022)	(362.791)	(84.235)	(346.132)
Mean Dep. Var. Treated	0.014	1724.233	2512.628	4236.860
Mean Dep. Var. Control	0.001	2466.218	637.964	3104.182
FE by Year	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓
Observations	558	564	564	564

Notes: This table shows the effects of the policy on the stock of firms. The sample comprises cohorts of all firms in exposed and non-exposed industries. Column (1) shows the likelihood of being subject to the Factor R rule. Column (2) shows the effects on the number of firms in *Simples Nacional*, while Column (3) uses the number of firms in *Lucro Presumido*. Column (4) calculates the net effect on the stock. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

Table A.6: Extensive-Margin Difference-in-Differences Estimates

	(1) # New Firms	(2) Log New Firms	(3) Sh. Employees t-1	(4) Sh. Priv. Employees t-1	(5) Sh. Top 5% Employees t-1
Pre-period					
2016	-25.244**	-0.112	0.005	-0.011	0.020***
	(10.820)	(0.092)	(0.020)	(0.016)	(0.006)
Post-period					
2018	129.444***	0.379***	-0.001	-0.004	-0.005
	(31.568)	(0.078)	(0.008)	(0.011)	(0.006)
2019	180.055***	0.452***	0.001	0.003	-0.006
	(46.114)	(0.069)	(0.008)	(0.011)	(0.007)
Mean Dep. Var. Treated	206.95	4.15	0.47	0.32	0.16
Mean Dep. Var. Control	208.47	4.05	0.37	0.28	0.07
FE by Year	✓	✓	✓	✓	✓
FE by Industry	✓	✓	✓	✓	✓
Reweighted	X	X	✓	✓	✓
Observations	564	472	466	466	466

Notes: This table reports event-study difference-in-differences estimates of the Factor R reform on extensive-margin outcomes. Each column corresponds to a separate regression. Coefficients are interactions of year indicators with treatment exposure, with 2017 as the omitted reference year. All specifications absorb year and industry (CNAE) fixed effects. Standard errors clustered at the industry level in parentheses. *Mean Dep. Var.* rows report the 2016 baseline mean of the dependent variable for treated and control groups. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

Table A.7: Anexo VI tax rates: pre- vs post-reform (Factor R)

Pre-reform

Annual Revenues (R$)	**Tax Rate**
180,000	16.93%
180,000–360,000	17.72%
360,000–540,000	18.43%
540,000–720,000	18.77%
720,000–900,000	19.04%
900,000–1,080,000	19.94%
1,080,000–1,260,000	20.34%
1,260,000–1,440,000	20.66%
…	…
3,420,000–3,600,000	22.45%

Post-reform

Annual Revenues (R$)	**Tax Rate**	
	FR < .28	**FR ≥ .28**
180,000	15.50%	6.00%
180,000–360,000	18.00%	11.20%
360,000–720,000	19.50%	13.50%
720,000–1,800,000	20.50%	16.00%
1,800,000–3,600,000	23.00%	21.00%
3,600,000–4,800,000	30.50%	33.00%

Notes: This Figure shows the tax schedule for firms in Anexo VI pre-reform, and when they become subject to the Factor R reform. Despite the top marginal revenue tax rate being larger post-reform when ≥ 28%, note that the last revenue bracket post-reform was not possible in the pre-reform period, as the *Simples Nacional* threshold was lower.

B Additional Context Information

Brazil is one of the largest middle-income countries with more than 220 million inhabitants. The production landscape is dominated by small and medium-sized firms, which accounted for more than 80% of the approximately 7 million active firms in 2022. SMEs are also important for employment as they comprise 31% of formal employees and 26% of total payroll. However, they represent a much smaller share of the government's revenue from business taxation (8%).

Table A.1 provides descriptive statistics for firms in each tax regime. The largest firms are in *Lucro Real*, a standard profit tax regime mandated for firms with annual revenues above R$78 million. The marginal tax rate on profits ranges from 15 to 35%. The second tax regime is *Lucro Presumido*, which also covers large firms as those with annual revenues above R$4.8 million must opt in either for *Lucro Presumido* or *Lucro Real*. The critical difference is that firms in this regime pay taxes on the presumptive profits. Common to both tax regimes is the requirement that firms submit several forms to the tax authority, including not only complete digital accounting records and fiscal bookkeeping but also state and municipal tax forms. SMEs fall under *Simples Nacional*, with a R$4.8 million threshold. This is a simplified tax regime, under which taxpayers submit a single, straightforward tax form to the tax authority to report basic information. SMEs are taxed on revenue, with the rate depending on revenue level and sector. Finally, low-income self-employed can opt-in for *Microemprendedor Individual* (MEI), as long as their annual revenues do not exceed R$81,000. This is also a simplified tax regime with a flat payment, and it is not legally allowed to have more than one employee. It was designed to increase registration of informal employees (Farias and Hsu Rocha, 2026).

This paper focuses on *Simples Nacional*, the cornerstone of SME policy. It was established in federal law in 2006, and it covers a heterogeneous range of firms, from high-income individuals providing consulting services to medium-sized labor-intensive manufacturing firms. Moreover, owners of firms in *Simples Nacional* rank between the 95th and 99.5th percentiles of the income distribution in the country (Palomo et al., 2025).

Within *Simples Nacional*, firms are currently assigned to one of the five sector-specific "Anexos" (schedules) with progressive marginal rates by gross-revenue brackets. The underlying principle of the Anexos is that different sectors have different profit margins, thereby making explicit the presumptive component of this simplified tax regime. Because the service sector is highly heterogeneous, Anexos III, IV, and V differentiate among activities within the service sector. Moreover, to further approximate true profit margins, some of these service-related activities are subject to the *Factor R* rule. This rule states that the corresponding Anexo (and therefore the tax schedule) depends on the firm's payroll-

to-revenue ratio. Firms with large Factor R are assigned to Anexos with lower revenue tax rates, and vice versa. Online Appendix J presents all tax schedules corresponding to *Simples Nacional*.

The tax regime of *Lucro Presumido* also deserves further explanation, as it is the main alternative to *Simples Nacional*, and, therefore, the key counterfactual we investigate. In this, firms pay corporate taxes on a statutory measure of presumed profits rather than on their actual profits. The presumed profit margin is a fixed share of gross revenue, depending on the firm's activity. For most commercial and industrial activities, the relevant margins are relatively low (8–12 percent of revenue) whereas for most service activities the presumed profit margin is 32 percent. Corporate taxes are then applied to these presumed profits, with a standard combined tax rate of 24 percent.[1] Payroll taxes and state or municipal taxes are assessed separately, as they are not part of a simplification regime. During our study period, firms could opt for Lucro Presumido if their previous-year revenues did not exceed R$78 million, subject to restrictions for certain activities and types of firms.

Lucro Presumido therefore differs substantially from Simples Nacional. First, its revenue ceiling is much higher: R$78 million compared with R$4.8 million under Simples Nacional. Second, while Lucro Presumido determines corporate income taxes by applying activity-specific presumed profit margins to revenue and requires firms to calculate the different taxes separately, Simples Nacional taxes firms more directly on gross revenue through a unified, progressive schedule whose rates depend on the firm's activity and revenue. Simples also consolidates most federal taxes, employer social-security contributions, and state or municipal taxes into a single payment, whereas these obligations are generally paid separately under Lucro Presumido. Thus, Simples provides both preferential taxation and substantial administrative simplification for small firms, while Lucro Presumido is primarily an alternative method of calculating taxable corporate profits and remains available to considerably larger firms.

C Conceptual Framework

C.1 Which firms gain from Factor R?

Equation 4 shows that heterogeneity in the value of Factor R can arise through two conceptually distinct mechanisms. Some firms naturally generate a sufficiently high payroll-to-revenue ratio and therefore receive the full benefit of the lower tax schedule. Other firms would naturally locate below the threshold but can profitably modify their behav-

[1]Corporate profit taxation consists of the *Imposto de Renda da Pessoa Jurídica* (IRPJ), levied at a standard rate of 15 percent, and the *Contribuição Social sobre o Lucro Líquido* (CSLL), generally levied at 9 percent.

ior to satisfy it. We refer to these mechanisms as *natural eligibility* and *costly eligibility*, respectively.

C.1.1 Natural eligibility and labor intensity

We first consider heterogeneity in firms' natural labor intensity. Recall that x_i^L denotes the allocation firm i would choose under the favorable tax schedule if access to that schedule did not depend on Factor R. Define its natural employee-payroll intensity as

$$p_i^N \equiv \frac{wL_i^L}{\widetilde{R}_i^L}, \tag{10}$$

and its natural owner-wage share as

$$\omega_i^N \equiv \frac{w_i^{o,L}}{\widetilde{R}_i^L}. \tag{11}$$

These are equilibrium objects resulting from the firm's production technology and its optimal choices under the low-tax schedule. In particular, p_i^N summarizes the heterogeneity in labor intensity. Then, the firm's natural Factor R can be written as

$$r_i^N = p_i^N + \omega_i^N. \tag{12}$$

A firm is *naturally eligible* if its unconstrained low-tax allocation already satisfies the statutory requirement,

$$r_i^N \geq T. \tag{13}$$

Therefore, a higher natural employee-payroll share makes natural eligibility more likely and may be sufficient to qualify for the low tax schedule whenever $p_i^N \geq T$.

Natural eligibility has an important implication for the reform's value. Because x_i^L already satisfies the eligibility requirement, imposing Factor R removes no allocation that the firm would otherwise choose. Thus, for naturally eligible firms, Factor R operates exactly as an unconstrained reduction in the revenue tax.

C.1.2 Costly eligibility

Consider instead a firm whose unconstrained low-tax allocation does not satisfy Factor R:

$$r_i^N < T. \tag{14}$$

Such a firm can obtain the favorable schedule only by departing from its preferred low-tax allocation. It may increase employee payroll, shift owner income from dividends to wages, reduce reported revenue, alter its production choices, or combine these margins. The optimized value loss from doing so is precisely Equation 3.

Costly eligibility is profitable if and only if

$$B > C. \tag{15}$$

To characterize how this comparison varies with potential scale Ai, define, for firms satisfying (14),

$$Q \equiv \frac{C}{B}. \tag{16}$$

Assumption 1 (Scale single crossing in costly eligibility)**.** *Fix the firm's non-scale technology F_i, and consider a connected interval of productivity levels A over which the firm does not naturally qualify:*

$$r_i^N(A) < T.$$

On this interval,

$$Q_i(A) \equiv \frac{C_i(A)}{B_i(A)}$$

is continuous and strictly increasing in A, and takes values both below and above one.

Assumption 1 requires the optimized cost of satisfying Factor R to increase with potential scale faster than the benefit of the favorable tax schedule. Importantly, all endogenous choices are allowed to adjust with A, including the firm's natural Factor R. Thus, the assumption does not compare firms at a fixed distance from the eligibility threshold. Instead, the assumption allows the firm's natural Factor R to change with scale, but it requires the scale force to dominate any endogenous change in the amount of adjustment required to qualify.

The intuition is especially transparent when the cheapest adjustment is owner-income shifting. Suppose, only for illustration, that a non-qualifying firm holds the other components of its unconstrained low-tax allocation fixed and closes the Factor R gap by reporting additional owner wages. The required additional owner wage is

$$m(A) = \left[T - r^N(A)\right] \widetilde{R}^L(A). \tag{17}$$

For a given distance from the threshold, the amount of income that must be shifted grows with firm scale. If the private cost of shifting owner income is convex, the cost of satisfying the same payroll-to-revenue requirement can therefore rise more than proportionally with scale. Assumption 1 requires this force to dominate any corresponding increase in the

value of the tax cut.

Proposition 1. *Natural and costly eligibility*

1. ***Natural eligibility.*** *If the firm's unconstrained low-tax allocation satisfies*

$$r_i^N \geq T, \tag{18}$$

then the eligibility restriction does not constrain the firm's preferred allocation. Therefore,

$$C_i = 0, \qquad G_i = B_i > 0. \tag{19}$$

In particular, $p_i^N \geq T$ is sufficient for natural eligibility.

2. ***Costly eligibility.*** *If the firm's unconstrained low-tax allocation satisfies $r_i^N(A) < T$ and Assumption 1 holds, there exists a unique cutoff $\overline{A}^M(F_i)$ (for a given non-scale technology) satisfying*

$$Q_i\left(\overline{A}^M(F_i)\right) = 1. \tag{20}$$

Within this non-natural-eligibility region,

$$A < \overline{A}^M(F_i) \quad \Longrightarrow \quad C_i(A) < B_i(A) \quad \Longrightarrow \quad G_i(A) > 0, \tag{21}$$

whereas

$$A > \overline{A}^M(F_i) \quad \Longrightarrow \quad C_i(A) > B_i(A) \quad \Longrightarrow \quad G_i(A) = 0. \tag{22}$$

At $A = \overline{A}^M(F_i)$, the firm is indifferent between satisfying Factor R and remaining under the high-tax schedule.

The proof is provided in Online Appendix G.

Proposition 1 identifies two distinct reasons why Factor R increases the value of Simples. First, firms whose unconstrained choices under the favorable tax schedule generate a sufficiently high Factor R are naturally eligible. For these firms, the eligibility requirement is irrelevant for behavior: they receive the full value of the lower tax schedule and choose exactly the allocation they would choose under an unconditional tax cut. Natural labor intensity is an important source of this heterogeneity, since firms with sufficiently high employee payroll relative to revenue may satisfy the requirement through employee payroll alone.

Second, firms whose natural Factor R lies below the threshold can benefit only by departing from their preferred low-tax allocation. Conditional on the firm's non-scale technology, Assumption 1 implies that costly eligibility is profitable at sufficiently low levels of potential scale.

C.2 Implications for incumbent firms

Consider firms that were already operating in Simples before the reform. For these firms, participation in the regime is held fixed, and the reform changes the allocation they choose within Simples.

To characterize the firm's post-reform location in the Factor R distribution, define its optimized payoff under the favorable schedule conditional on attaining a particular Factor R:

$$W_{L,i}(r) \equiv \max_{x_i : r(x_i)=r} u_L(x_i). \tag{23}$$

By construction, $W_{L,i}(r)$ reaches its unconstrained maximum at the firm's natural Factor R, r_i^N. We impose the following regularity condition.

Assumption 2. *Factor R single-peakedness The set of attainable Factor R values is an interval containing T, and $W_{L,i}(r)$ is continuous and strictly single-peaked at r_i^N.*

The relevant implication of Assumption 2 is that when $r_i^N < T$, the firm's best allocation among those satisfying $r_i \geq T$ lies exactly at the eligibility threshold. Once the firm moves above T, moving farther above it provides no additional tax benefit and only moves the firm farther from its preferred low-tax allocation.

Let r_i^0 denote the firm's optimal Factor R before the reform, when it faces the high-tax schedule, and let r_i^1 denote its optimal Factor R after the reform.

Proposition 2 (Incumbent responses around the Factor R threshold)**.** *Under Assumption 2:*

1. ***Initially above the threshold.*** *If $r_i^0 \geq T$, the firm never crosses below the threshold and*

$$r_i^1 = \begin{cases} r_i^N, & \text{if } r_i^N \geq T, \\ T, & \text{if } r_i^N < T. \end{cases}$$

2. ***Initially below the threshold.*** *If $r_i^0 < T$, then*

$$r_i^1 = \begin{cases} r_i^N, & \text{if } r_i^N \geq T, \\ T, & \text{if } r_i^N < T \text{ and } B_i > C_i, \\ r_i^0, & \text{if } r_i^N < T \text{ and } B_i < C_i. \end{cases}$$

When $r_i^N < T$ and $B_i = C_i$, the firm is indifferent between remaining below the threshold and bunching at T.

The proof is provided in Online Appendix G.

The first part of Proposition 2 delivers a particularly clean prediction. A firm that was already above the threshold before the reform never optimally crosses below it. The reason does not depend on the firm's technology or on how its inputs respond to the tax cut. The firm can always retain its pre-reform allocation, remain eligible, and pay strictly lower taxes at that same allocation. Hence, some qualifying allocation is strictly preferable to any allocation attainable under the high-tax schedule.

For firms initially below the threshold, obtaining the favorable schedule instead requires an upward movement in Factor R. If the firm's unconstrained low-tax optimum already satisfies $r_i^N \geq T$, it crosses the threshold without being constrained once it does so. If $r_i^N < T$, however, qualifying requires the firm to depart from its preferred low-tax allocation. Proposition 1 determines whether this costly eligibility is worthwhile. A firm for which $B_i > C_i$ qualifies, and because locating farther above the threshold provides no additional discrete tax benefit, it bunches at T. Otherwise, it remains below the threshold.

An important implication is that the excess mass at the threshold can originate from both sides of the pre-reform Factor R distribution. Firms initially below T can move upward to obtain the favorable schedule, while firms initially above T can re-optimize downward until the eligibility condition binds. Proposition 2 rules out only one movement: an initially above-threshold firm should not cross below the eligibility threshold.

The framework also provides a direct mapping to observable adjustment margins. Let

$$P_i = wL_i + w_i^o.$$

Any firm locating at the threshold after the reform satisfies

$$P_i^1 = T\widetilde{R}_i^1.$$

Since prior to the reform

$$P_i^0 = r_i^0 \widetilde{R}_i^0,$$

changes between the pre- and post-reform allocations satisfy the accounting identity

$$\Delta P_i - T\Delta\widetilde{R}_i = (T - r_i^0)\widetilde{R}_i^0. \tag{24}$$

For a buncher coming from below, $r_i^0 < T$, so

$$w\Delta L_i + \Delta w_i^o - T\Delta\widetilde{R}_i = (T - r_i^0)\widetilde{R}_i^0 > 0. \tag{25}$$

The firm must therefore close its initial distance from the threshold through some com-

bination of higher employee payroll, higher owner wages, or lower reported revenue. Which margin moves depends on the relative economic costs of the available responses. Their observed changes therefore reveal how firms find it cheapest to satisfy Factor R.

For a buncher coming from above, the right-hand side of (24) is negative. Such a firm would like to reduce its payroll-to-revenue ratio in response to the favorable schedule, but the eligibility restriction prevents it from moving below T.

C.3 Implications for selection into Simples

We next consider potential firms choosing among Simples Nacional (S), Lucro Presumido (P), and an outside option (0). The outside option includes both inactivity and informality, as we can't empirically identify among them.

Let K_S denote the fixed cost of operating under Simples Nacional, including registration and compliance costs. Pre-reform, the value of choosing S for a given firm is:

$$V_S^0 = \Pi_H - K_S, \tag{26}$$

while after the reform,

$$\begin{aligned} V_S^1 &= \max\{\Pi_H, \Pi_T\} - K_S \\ &= V_S^0 + G. \end{aligned} \tag{27}$$

Equation (27) provides the direct connection between the intensive and extensive margins. Factor R raises the value of selecting into S by exactly the same object G that governs the incentive to obtain the favorable schedule for firms already operating in the regime.

Proposition 1 therefore has an immediate selection implication. Factor R makes S more valuable for both firms that naturally satisfy the eligibility requirement and firms that can reach eligibility at sufficiently low cost. Whether this increase in value actually induces a firm to select into S depends on its alternative regime choice.

Let V_P denote the value of operating under Lucro Presumido, which is unaffected by Factor R, and normalize the outside option to zero. To characterize the counterfactual regime of firms induced into Simples, we impose the following productivity-sorting condition.

Assumption 3 (Productivity sorting across regimes)**.** *Fix the firm's non-scale technology F_i.*

1. *$V_S^0(A)$ is continuous and strictly increasing in A and crosses the outside option exactly once. Let $A_E^0(F_i)$ satisfy*

$$V_S^0(A_E^0(F_i)) = 0.$$

2. *The relative value of Lucro Presumido,*

$$D^0 \equiv V_P - V_S^0,$$

is continuous and strictly increasing in A and crosses zero exactly once at $A_{SP}^0(F_i)$, with

$$A_E^0(F_i) < A_{SP}^0(F_i).$$

The first condition captures the standard scale effect in formal entry: sufficiently low-productivity firms do not generate enough surplus to cover the fixed costs of formal operation. The second is a single-crossing condition between the two formal tax regimes. It states that Lucro Presumido becomes relatively more attractive as productivity and potential scale increase. A natural microfoundation is that Lucro Presumido entails higher fixed accounting and compliance costs than Simples but offers a relatively more attractive variable payoff at larger scale.

Assumption 3 implies the following pre-reform allocation, apart from exact indifference points:

$$j_i^0 = \begin{cases} 0, & A_i < A_E^0(F_i), \\ S, & A_E^0(F_i) < A_i < A_{SP}^0(F_i), \\ P, & A_i > A_{SP}^0(F_i). \end{cases} \tag{28}$$

Thus, conditional on the firm's non-scale technology, potential scale orders firms from the outside option, to Simples, to Lucro Presumido.

Proposition 3 (Counterfactual regime of firms induced into Simples)**.** *Under Assumption 3, fix the firm's non-scale technology F_i and consider a firm induced by Factor R to choose Simples.*

1. *If its pre-reform counterfactual is the outside option, then*

$$A_i < A_E^0(F_i).$$

2. *If its pre-reform counterfactual is Lucro Presumido, then*

$$A_i > A_{SP}^0(F_i).$$

Since $A_E^0(F_i) < A_{SP}^0(F_i)$, conditional on the same non-scale technology, firms induced from Lucro Presumido have higher potential scale than firms induced from the outside option.

The proof is provided in Online Appendix G.

The same object G determines whether Factor R expands Simples at either margin. If Factor R raises the value of Simples for firms at the pre-reform outside–Simples boundary, the lower boundary shifts downward, inducing firms that would otherwise choose the outside option to select into Simples. Similarly, if it raises the value of Simples at the pre-reform Simples–Lucro Presumido boundary, the upper boundary shifts upward, inducing firms that would otherwise operate under Lucro Presumido. These boundary movements require the corresponding single-crossing properties to continue to hold after the reform; we state the precise conditions in Online Appendix G.

Propositions 1 and 3 therefore characterize two distinct dimensions of selection. Proposition 1 determines *why* Factor R raises the value of Simples: because the firm naturally satisfies the eligibility rule or because it can reach eligibility at sufficiently low cost. Proposition 3 instead determines *where* an induced firm would have operated in the absence of the reform.

Importantly, the framework does not impose a one-to-one mapping between these two dimensions. Firms induced from either the outside option or Lucro Presumido may benefit through natural or costly eligibility. In particular, although costly eligibility becomes less attractive with potential scale conditional on a given non-scale technology, the scale thresholds governing eligibility and regime choice both depend on F_i. Whether marginal entrants are selected primarily because of their natural labor intensity or because of their ability to reach eligibility, and whether these firms would otherwise have chosen the outside option or Lucro Presumido, are therefore empirical questions.

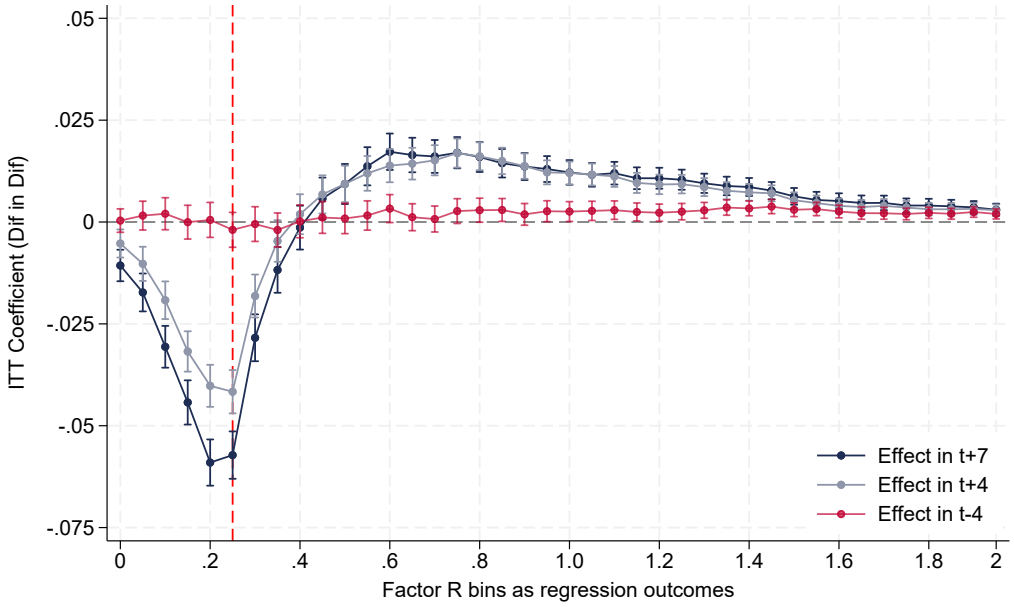
D Location in the Factor R distribution for incumbent firms

Proposition 2 shows that the excess mass at the 28% threshold can originate from *both* sides of the pre-policy Factor R distribution: firms initially below 28% may move up to obtain the tax reduction, while firms initially above 28% may re-optimize downward toward the threshold. This appendix provides an empirical test of the proposition: we estimate where the firms that end up at 28% would have been located absent the reform, and we quantify how much of the bunching is drawn from each side of the threshold.

To do so, we estimate the difference-in-differences specification in Equation (7) using the firm's *location* in the Factor R distribution as the outcome. We discretize the Factor R support into bins and, for each bin cutoff c, define the outcome $y_{iq} = \mathbf{1}[FR_{iq} \leq c]$, an indicator for whether firm i's Factor R in quarter q lies at or below c. This outcome is the empirical cumulative distribution function (CDF) evaluated at c, so estimating Equation (7) bin by bin traces the treatment effect on the *entire* distribution of Factor R rather than on a single moment. Figure D.7 summarizes these estimates. Each point on the horizontal

axis corresponds to a different regression, that is a different outcome $\mathbf{1}[FR_{iq} \leq c]$ indexed by the bin cutoff c, and the vertical axis reports the associated difference-in-differences coefficient β_k, the treatment effect on the share of firms located at or below that bin. Each line collects the β_k for a given event time k: we plot the pre-reform placebo at $t-4$ together with the post-reform effects at $t+4$ and $t+7$. A negative coefficient at cutoff c means the reform reduced the mass of treated firms located at or below c; a positive coefficient means it raised it.

Figure D.7: Effect on the Factor R Cumulative Distribution Function



Notes: This figure shows the reduced form coefficients (β_ks) from Equation 7 where the outcome $y_{iq} = \mathbf{1}[FR_{iq} \leq c]$ and c corresponds to each bin in the Factor R distribution. Then, each dot in the x-axis refers to a different regression. Bins are of size .05, with the first bin including the zero, and the last bin comprising any value above 2 to cover the entire population of firms. The lines plot different ks: the red line is the dif-in-dif coefficient in -4, while the blues are post-period coefficients.

Two features of the dynamics are worth noting. First, the pre-reform line ($t-4$) is flat and statistically indistinguishable from zero across the entire distribution, confirming that treated and control firms followed parallel paths before the policy and that the estimated effects are not driven by pre-existing differences in where firms were located. Second, the response builds up over time: the displacement of mass toward the threshold is already visible at $t+4$ but is substantially larger at $t+7$, consistent with the descriptive evidence that bunching grows year after year as incumbent firms learn to exploit the notch. We end the event window at $t+7$, the fourth quarter of 2019, for two reasons. It stops the analysis before the COVID-19 shock of 2020, which would otherwise confound the comparison of treated and control firms; and it keeps the incumbent sample large, since requiring firms to be continuously active over a longer window would progressively drop firms and erode the set of incumbents we can follow.

The shape of the curve identifies where the bunching mass comes from. Because the plotted outcome is the CDF, its slope across bins is the treatment effect on the *density* (the PDF): where the curve rises the reform pushed mass into that region, and where it falls the reform pulled mass out. The bunching at 28% therefore appears as the steep positive slope at the threshold. Reading the $t+7$ line, the effect on the CDF turns increasingly negative as we approach the threshold from below and reaches its trough just at 28%: the reform emptied the region immediately below the threshold, as firms that would have located there moved up to bunch. Crucially, just above the threshold the curve does not return to zero but turns *positive*. This is only possible if part of the bunching is composed of firms that would otherwise have been located higher up and re-optimized downward toward the threshold.

This logic delivers a clean decomposition of the bunching mass by origin. Since treated and control firms are both distributions, total mass is conserved and the coefficient curve begins and ends at zero; the mass added by the positive slope at the threshold must therefore be exactly offset by the mass removed elsewhere. Two regions of the curve have negative slope, and hence correspond to mass being removed. Because the negative slopes must sum to the positive slope, we can attribute the bunching to each side. We find that 84% of the firms that bunch at 28% would have been located below the threshold, while the remaining 16% would have been located above it.

E Revenue Elasticity

Figure 3b shows that High-FR firms increase the reported revenues in response to the automatic eligibility for the revenue tax cut. To translate the reduced form result into a revenue elasticity, we need to account for the fact that the revenue tax rate depends on the revenue bracket, which may change due to the reform. We do so by instrumenting the revenue tax rate with treatment status and performing 2SLS on the log of revenues. Note that we are forced to use the log of revenues rather than a Poisson specification to avoid the *forbidden regression* in 2SLS. However, as our sample of incumbent firms comprises firms with positive revenues in all periods, we are not affecting the sample and the reduced form estimates are virtually equivalent for the particular outcome of revenues.

We define the elasticity to the net-of-tax rate between t and $t-1$ as:

$$\varepsilon_{R,1-\tau} = \frac{\mathbb{E}\left[\log\left(\frac{R_{i,t}}{R_{i,-1}}\right) \mid \text{Treated} = 1\right] - \mathbb{E}\left[\log\left(\frac{R_{i,t}}{R_{i,-1}}\right) \mid \text{Treated} = 0\right]}{\mathbb{E}\left[\log\left(\frac{1-\tau_{i,t}}{1-\tau_{i,-1}}\right) \mid \text{Treated} = 1\right] - \mathbb{E}\left[\log\left(\frac{1-\tau_{i,t}}{1-\tau_{i,-1}}\right) \mid \text{Treated} = 0\right]} \tag{29}$$

Where the choice of t recovers different time-horizon elasticities. The numerator captures the firm's response, and the denominator measures the size of the incentives. Equa-

tion 30 shows the 2SLS equations that we estimate to recover Equation 29. The underlying identifying assumption is that the treated and control groups would follow parallel trends in the absence of the policy.

$$\begin{aligned} \Delta_{-1,t} \ln y_i &= \alpha_0 + \varepsilon_{R,t} \Delta_{-1,t} \ln (1 - \tau_i) + u_{it} \\ \Delta_{-1,t} \ln (1 - \tau_i) &= \gamma_0 + \pi \mathbf{1}[\text{Anexo VI} = 1] + v_{it} \end{aligned} \tag{30}$$

We find sizable revenue elasticities with respect to the revenue tax rate. Table E.8 shows that two years after the policy, High FR firms have increased reported revenues by 1% for each 1% increase in the net-of-tax rate. Moreover, consistent with the previous patterns, firms seem to be gradually responding as the coefficients grow over time.

Table E.8: Revenue Elasticity Estimates Over Time Horizons (High FR)

	12-month Revenue		Quarter Revenue	
	1 Year	2 Year	1 Year	2 Year
ε_R	.392	.887	.994	1.374
SE	(0.045)	(0.063)	(0.067)	(0.084)
CI	[0.30, 0.48]	[0.76, 1.01]	[0.86, 1.13]	[1.21, 1.54]
Reduced Form	.040	.089	0.101	0.137
First Stage (γ_1)	0.101	0.100	0.101	0.100
Observations	34,014	34,014	34,014	34,014

Notes: This table shows the coefficients estimated from Equation 30 for different time horizons. Columns 1 and 2 use as outcome the log of 12-month accumulated revenues (Factor R denominator). Columns 3 and 4 use the monthly average of the last quarter of 2018 and 2019, respectively. The sample contains only firms in the High FR group.

F Details for Revenue Implications

This section provides the details behind the revenue calculations reported in Section 6. We focus on the medium-term effect of the reform and evaluate behavioral responses using the last quarter of 2019, the latest period in our event study. This is useful because the behavioral responses build gradually following the reform.

One quarter before the policy's implementation, there were about $N^{treat} = 96,000$ existing firms in activities affected by the reform.[2] Based on their pre-reform Factor R, 30%

[2] N^{treat} = 96,000 counts all firms with positive revenue in 2017Q4 in exposed activities, not only the balanced panel of Section 4; the latter requires survival through 2019Q4, which is not the relevant population for the fiscal calculation.

belong to the High-FR group and 70% to the Low-FR group. Average 12-month reported revenues in 2017Q4 are R$220,681 among High-FR firms and R$308,565 among Low-FR firms, as reported in Table 1.

Mechanical Effect

The mechanical effect holds firms' reported revenues and all other behavior fixed at their pre-reform levels. High-FR firms were already above the 28 percent threshold and therefore became automatically eligible for Anexo III after the reform. The mechanical change in revenue collection is consequently:

$$\Delta Rev^{mech} = -(\tau_{VI} - \tau_{III})R^{hi}.$$

Where $(\tau_{VI} - \tau_{III})$ is the difference in the tax rates induced by the reform. R^{hi} is the total revenue of firms mechanically affected by the reform which equals the share of firms above 28% pre-reform times their average revenues. [3] Using the values above, ΔRev^{mech} = –R$ 572 millions.

Incumbent Firms: reported-revenue increase of High-FR firms

High-FR firms subsequently increase reported revenues in response to the lower revenue tax rate. For the medium-term calculation, we use the reduced-form effect in 2019Q4, the last period in our event study. Table E.8 reports an estimated increase in quarterly reported revenue of $\hat{\rho}_{2019Q4}$ = .137. This differs from the accumulated 12-month revenue because the latter averages together earlier quarters in which firms had not yet fully adjusted. Therefore, $\hat{\rho}_{2019Q4}$ better captures the revenue response prevailing by the end of our estimation period.

As a result, $\Delta Rev^{R} = \tau_{III} \times \hat{\rho}_{2019Q4} \times R^{hi}$ = R$ 52 millions denotes the increase in revenue collection resulting from the incumbent firms' revenue response.

Incumbent Firms: Income-shifting response of Low-FR firms

Low-FR firms are not automatically eligible for the tax reduction. The relevant revenue effect therefore depends on both how many firms are induced to cross the 28% threshold and the pre-reform revenue of those firms. Because the conceptual framework predicts that the profitability of costly eligibility may vary systematically with firm scale, we do not assume that induced crossers have the average revenue of all Low-FR incumbents. Instead, we estimate these objects directly using the Low-FR incumbent sample.

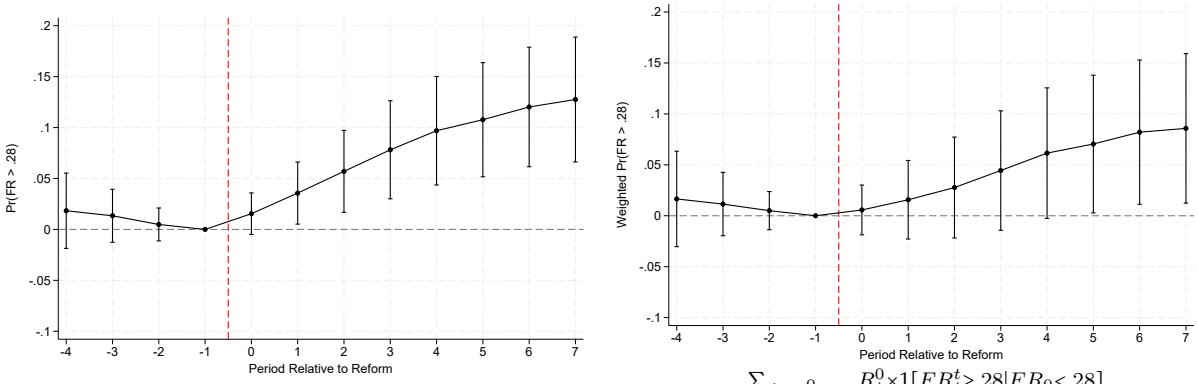
Let $\hat{c}$ denote the fraction of initially Low-FR firms induced by the reform to locate above the eligibility threshold. We can estimate $\hat{c}$ using $\Pr(FR_{i,2019Q4} \geq .28 | FR_{i,2017Q4} < .28)$

[3] $R^{hi} = 96,000 \times 0.30 \times 220,681 = R\$6.356$ billions.

as the outcome variable in Equation 7.

Second, to recover the revenue base of the firms induced to cross, let R_i^0 denote the firm's pre-reform 12-month revenue in 2017Q4, and estimate the same specification using $Y_{i,2019Q4}^{R,C} = R_i^0 \times \Pr(FR_{i,2019Q4} \geq .28 | FR_{i,2017Q4} < .28)$. Let's call $\hat{m}$ the resulting coefficient. Then $\hat{m}$ measures the pre-reform revenue mass of induced crossers per initially Low-FR firm.

Figure F.8: Treatment vs. Control (Low Factor R pre-reform)



(a) $\Pr(FR_{i,t} \geq .28 | FR_{i,2017Q4} < .28)$

(b) $\frac{\sum_{i|FR_i^0<.28} R_i^0 \times 1[FR_i^t \geq .28 | FR_0 < .28]}{\sum_{i|FR_i^0<.28} R_i^0}$

Notes: This Figure shows the coefficients from Equation 7 using a Factor R of at least 28% as the outcome. It restricts the sample to those initially below 28% in order to recover the share of income-shifters (Panel a) and the weighted share of income-shifters (Panel b). The latter weights by firms' revenues, meaning it recovers the share of revenues that becomes taxed under the low rate due to the income-shifting response.

Therefore, the change in revenue collection due to income-shifting can be expressed as:

$$\Delta Rev^{cross} = -(\tau_V - \tau_{III}^P) N^{lo} \hat{m},$$

where τ_{III}^P takes into account that crossing is achieved through income-shifting, implying payroll tax payments on 28% of the reported revenues.[4]

Extensive Margin

To account for the extensive margin in the revenue calculation, we use the reform's effect on the stock of firms, rather than only on new entrants. While the new entrants analysis cleanly illustrates selection into *Simples Nacional*, it misses several extensive-margin responses that matter for revenue. First, new firms that register in year t, but only made positive revenues in $t+1$ do not show up in the definition of new entrants used in Table 3. Second, existing firms in *Lucro Presumido* may decide to switch to *Simples Nacional*, affect-

[4] As most crossers were already reporting some owner wages, adding payroll taxes for the entire 28% of revenues leads to an overestimate of the counterfactual payments and, therefore, a lower bound of the impact on the government's budget.

ing tax payments and the stock of firms in both regimes. Third, firms in *Simples Nacional* that would have graduated to *Lucro Presumido* may decide not to do so. Finally, firms in *Simples Nacional* may become less likely to close, affecting the exit margin.

The stock estimates in Table A.5, therefore, summarize the cumulative effect of these margins on the number of firms operating under each tax regime at a given point in time. For the medium-term revenue calculation, we use the 2019 estimates. Since the unit of observation in these regressions is the CNAE-year, the coefficients measure the change in the number of firms per exposed CNAE relative to non-exposed CNAEs. We aggregate these reduced-form estimates across the 86 CNAEs classified as exposed to the reform.

Counterfactual: no registration

We first consider the part of the increase in the stock of firms in *Simples Nacional* whose counterfactual is not revenue registration. To isolate this margin, we use the combined stock of firms in *Simples Nacional* and *Lucro Presumido*, reported in column (4) of Table A.5. Reallocation between the two tax regimes has no effect on this outcome: a firm moving from *Lucro Presumido* to *Simples Nacional*, for example, increases the stock of the latter by one and reduces the stock of the former by one. The effect on their combined stock therefore isolates the net increase in firms operating in these two formal tax regimes.[5]

In 2019, the estimated effect on the combined stock is

$$\widehat{\beta}_{2019}^{Total} = 536.077$$

firms per exposed CNAE. We therefore estimate the number of additional registered firms whose counterfactual is the outside option as

$$\widehat{N}^0 = N^{CNAE}\widehat{\beta}_{2019}^{Total} = 86 \times 536.077 = 46,103.$$

This object incorporates not only additional entry, but also any policy-induced reduction in exit that increases the stock of firms in the two regimes. Let $\bar{R}^0$ denote the average annual reported revenue used to value these additional firms. Since these firms operate in *Simples Nacional* under the reform and would not have generated business-tax revenue in the counterfactual, their contribution to government revenue is

$$\Delta Rev^0 = \tau_{III}^P \bar{R}^0 \widehat{N}^0,$$

where τ_{III}^P is the effective tax rate under the favorable Factor R schedule, including the tax cost associated with the owner-wage reporting used to satisfy the eligibility requirement.

[5] Table A.6 and Figure A.6 rule out alternative counterfactual.

Counterfactual: Lucro Presumido tax regime

We next consider firms that operate in *Simples Nacional* because of the reform but whose counterfactual is *Lucro Presumido*. Column (3) of Table A.5 directly captures this displacement margin. In 2019, the reform reduces the stock of firms in *Lucro Presumido* by

$$\widehat{\beta}^{LP}_{2019} = -303.281$$

firms per exposed CNAE. Thus, the estimated number of firms displaced from *Lucro Presumido* is

$$\widehat{N}^{LP} = -N^{CNAE}\widehat{\beta}^{LP}_{2019} = 86 \times 303.281 = 26,082.$$

Notice that the two counterfactuals exactly account for the estimated increase in the stock of firms in *Simples Nacional*:

$$839.358 = 536.077 + 303.281.$$

The first component represents an increase in the combined stock of formal firms, whereas the second represents reallocation from *Lucro Presumido* to *Simples Nacional*.

For firms displaced from *Lucro Presumido*, the fiscal cost is not the full tax liability paid under *Simples Nacional*, since these firms would have paid taxes under *Lucro Presumido* in the absence of the reform. Let $\bar{R}^{LP}$ denote their average annual reported revenue. Their contribution to the revenue effect is therefore

$$\Delta Rev^{LP} = -(\tau_{LP} - \tau^{P}_{III})\bar{R}^{LP}\widehat{N}^{LP}.$$

Thus, this component captures only the change in tax payments generated by reallocating a firm from *Lucro Presumido* to the favorable *Simples Nacional* schedule.